PHYSICS

1) A new scale of temperature (which is linear) called the W scale, the freezing and boiling points of water are 39°W and 239°W respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale ?

(1) 200° W
(2) 139° W
(3) 78° W
(4) 117° W

2) If a system is in thermal equilibrium with its surrounding, it means :

(1) Temperature of system & surrounding must be same
(2) Pressure, Volume, Temperature of system & surrounding must be same
(3) Pressure, Volume, Temperature of system & surrounding may be different
(4) None

3) The coefficients of apparent volume expansion of a liquid, when determined using two different vessels A and B are γ_1 and γ_2 respectively. If the coefficient of linear expansion of the vessel A is α, the coefficient of linear expansion of the vessel B is :-

(1) $\frac{\alpha\gamma_1\gamma_2}{\gamma_1+\gamma_2}$
(2) $\frac{\gamma_1-\gamma_2}{2\alpha}$
(3) $\frac{\gamma_1-\gamma_2+\alpha}{3}$
(4) $\frac{\gamma_1-\gamma_2}{3}+\alpha$

4) The co-efficient of linear expansion of crystal in one direction is α_1 and in other two directions is α_2 then co-efficient of volume expansion will be :-

(1) $2\alpha_1 + \alpha_2$
(2) $3\alpha_1$
(3) $\alpha_1 + 2\alpha_2$
(4) $3\left(\frac{\alpha_1+\alpha_2}{2}\right)$

5) A pendulum clock is 5 sec. fast at temperature 15°C and 10 sec. slow at 30°C, find the temperature at which clock gives correct time:-

(1) 25°C

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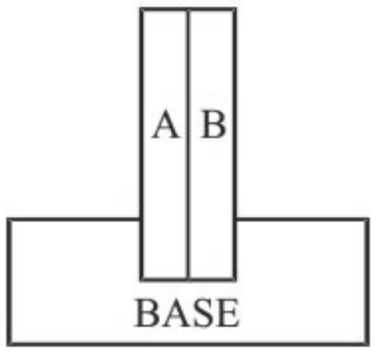
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(2) 23°C

(3) 20°C

(4) 18°C

6) A bimetallic strip consists of metals A and B. It is mounted rigidly at the base as shown. The metal A has a higher coefficient of expansion to that for metal B. When bimetallic strip is placed in a cold bath it will :-



(1) Bend towards the right

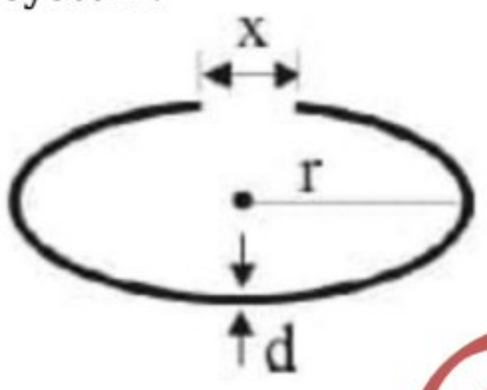
(2) Bend towards the left

(3) Not bend but shrink

(4) Neither bend nor shrink

7)

A cylindrical metal rod is shaped into a ring (radius r) with a small gap as shown. On heating the system:



(1) x decreases, r and d increases

(2) x and r increases, d decreases

(3) x, r and d all increase

(4) x and r decreased, d remins constant

8) The water equivalent of 20 gm. of aluminium (specific heat 0.2 cal/gm °C) is:-

(1) 40g

(2) 4g

(3) 8g

(4) 160g

9) Find out amount of heat required to convert 2 gm ice at –10°C into water at 60°C.

(1) 130 cal

(2) 290 cal

(3) 300 cal

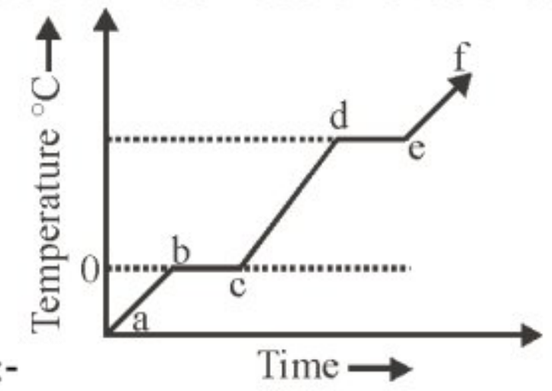
(4) 400 cal

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10) The following figure represents the temperature versus time plot for a given amount of a substance when heat energy is supplied to it at a fixed rate and at a constant pressure . Which part of the above plot represent a phase change :-



(1) a to b and e to f

(2) b to c and c to d

(3) d to e and e to f

(4) b to c and d to e

11) A liquid of specific heat 0.8 cal/g°C at temperature 60°C is mixed with other liquid of same mass having temperature 45°C. If temperature of mixture is 53°C, then specific heat (in cal/g°C) of the liquid is :-

(1) 0.5

(2) 0.6

(3) 0.7

(4) 0.8

12) A block of ice with mass m falls into a lake. After impact, a mass of ice m/5 melts. Both the block of ice and the lake have a temperature of 0°C. If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is

(1) $\frac{L}{5g}$

(2) $\frac{5L}{g}$

(3) $\frac{gL}{5m}$

(4) $\frac{mL}{5g}$

13) A piece of ice (heat capacity = 2100 J kg^{-1} $°C^{-1}$ and latent heat = 3.36×10^5 J kg^{-1}) of mass m grams is at –5°C at atmospheric pressure. It is given 420 J of heat so that the ice starts melting. Finally when the ice-water mixture is in equilibrium, it is found that 1 g of ice has melted. Assuming there is no other heat exchange in the process, the value of m is

(1) 40 g

(2) 8 g

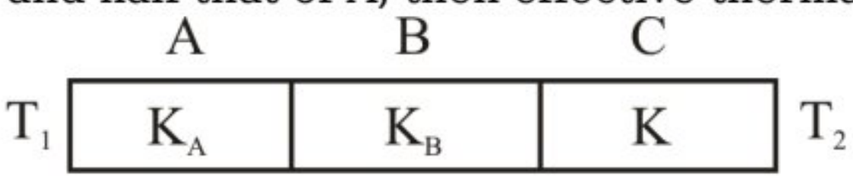
(3) 16 g

(4) 24 g

14) Three identical metal rods in dimension A, B and C are placed end to end and a temperature difference is maintained between the ends of A and C. If thermal conductivity of B is thrice that of C

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and half that of A, then effective thermal conductivity of the system is :-



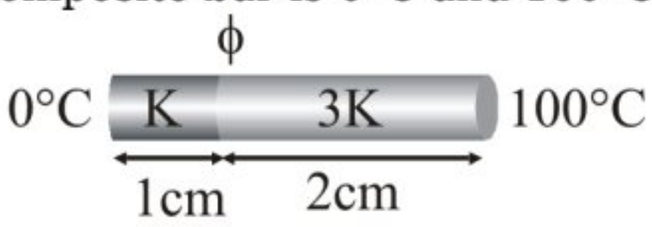
(1) 12K/7

(2) 6K/7

(3) 7K/3

(4) 2K

15) Two bars of thermal conductivites K and 3K and lengths 1 cm and 2 cm respectively have equal cross-sectional area, they are joined lengths wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature ϕ of the interface is :-



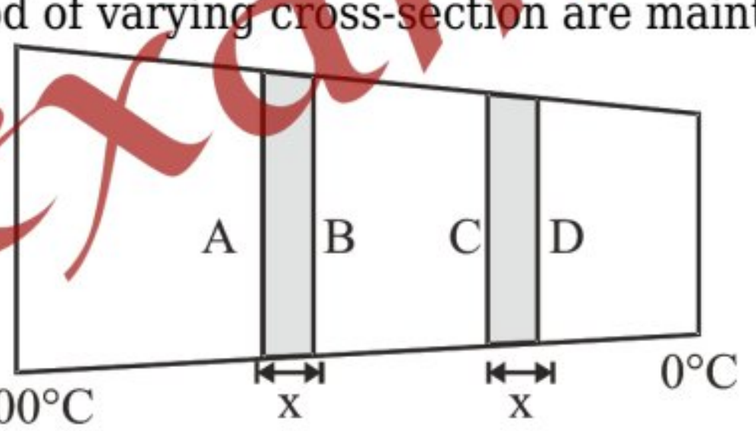
(1) 50°C

(2) $\frac{100^\circ}{3}$ C

(3) 60°C

(4) $\frac{200^\circ}{3}$ C

16) Two ends of a conducting rod of varying cross-section are maintained at 200°C and 0°C respectively. In steady state :-



(1) temperature differences across AB and CD are equal

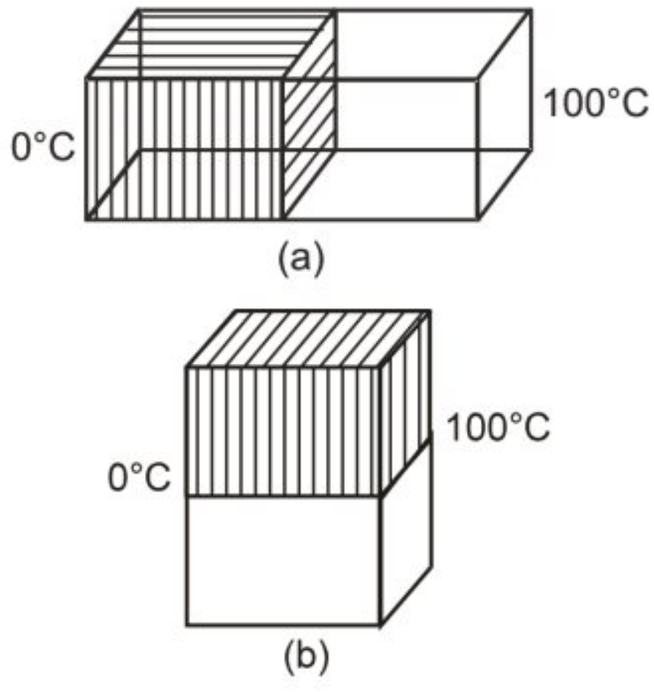
(2) temperature difference across AB is greater than that across CD

(3) temperature difference across AB is less than that across CD

(4) temperature difference may be equal or different depending upon thermal conductivity of the rod

17) Two identical square rods of metal are welded end to end as shown in figure (a). Assume that 10 cal of heat flows through the rods in 2 min. Now the rods are welded as shown in figure, (b). The

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time it would take for 10 cal to flow through the rods now, is

(1) 0.75 min

(2) 0.5 min

(3) 1.5 min

(4) 1 min

18) Ice starts forming in lake with water at 0ºC and when the atmospheric temperature is –10ºC. If the time taken for formation 1 cm of ice be 7 hours, then the time taken for the thickness of ice to change from 1 cm to 2 cm is :

(1) 7 hours

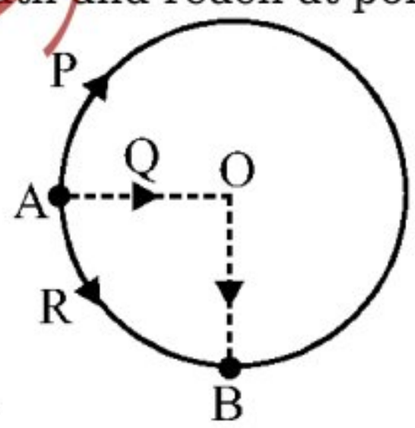
(2) 14 hours

(3) 21 hours

(4) 28 hours

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19) Three particles P, Q and R are situated at point A on the circular path of radius 10 m. All three particles move along different path and reach at point B shown in figure. Then the ratio of distance traversed by particle R and Q is:-



(1) $\frac{3}{4}$

(2) $\frac{1}{3}$

(3) $\frac{3\pi}{4}$

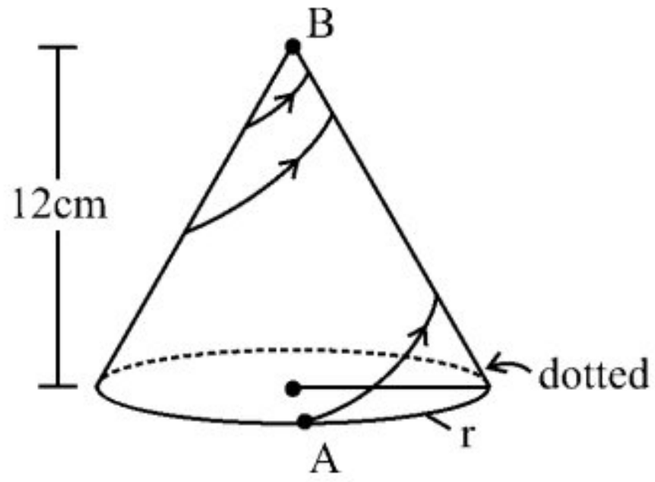
(4) $\frac{\pi}{4}$

20) An insect starts climbing a conical birthday hat of radius 5 cm at base. It starts from point A and reaches point B, taking spiral path on the hat. Find out its displacement if height is 12 cm:-

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(1) 12 cm

(2) 8 cm

(3) 13 cm

(4) 25 cm

21) If a car covers $\left(\frac{2}{5}\right)^{th}$ of the total distance with speed v_1 and $\left(\frac{3}{5}\right)^{th}$ distance with speed v_2 then average speed is

(1) $\sqrt{v_1 v_2}$

(2) $\frac{v_1 v_2}{2}$

(3) $\frac{5v_1 v_2}{2v_1 + 3v_2}$

(4) $\frac{5v_1 v_2}{3v_1 + 2v_2}$

22) A particle moving in a straight line covers half the distance with speed of 3 m/s. The other half of the distance is covered in two equal time intervals with speed of 4.5 m/s and 7.5 m/s respectively. The average speed of the particle during this motion is :-

(1) 4.0 m/s

(2) 5.0 m/s

(3) 5.5 m/s

(4) 4.8 m/s

23) The displacement of a particle is represented by the following equation : $s = 3t^3 + 7t^2 + 5t + 8$ where s is in metres and t in seconds. The acceleration of the particle at t = 1s is :–

(1) 14 m/s^2

(2) 18 m/s^2

(3) 32 m/s^2

(4) zero

24) A body thrown vertically upwards with an initial velocity u reaches maximum height in 4 seconds. The ratio of the distance travelled by the body in the first second and the seventh second is :-

(1) 3 : 2

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(2) 5 : 3

(3) 1 : 5

(4) 7 : 5

25) The velocity v of a moving particle varies with displacement as $x = \sqrt{v+1}$, the acceleration of the particle at x = 5 unit will be :-

(1) $\sqrt{6}$ unit

(2) 24 unit

(3) 240 unit

(4) 25 unit

26) Starting from rest, the acceleration of a particle is a = 2(t–1). The velocity of the particle at t=5 s is:-

(1) 15 m/s

(2) 25 m/s

(3) 5 m/s

(4) None of these

27) Position of a particle moving along x–axis is given by $x = (8t - 4t^2)$ m. The distance travelled by the particle from t = 0s to t = 2s is :-

(1) 0

(2) 8 m

(3) 12 m

(4) 16 m

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28) A bullet from a gun is fired on a rectangular wooden block with velocity *u*. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$ Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is :

(1) 24 cm

(2) 28 cm

(3) 30 cm

(4) 27 cm

29) A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β and comes to rest. If the total time elapsed is t, the maximum velocity acquired by the car will be:

(1) $\frac{\alpha^2 - \beta^2}{\alpha\beta}t$

(2) $\frac{\alpha^2 + \beta^2}{\alpha\beta}t$

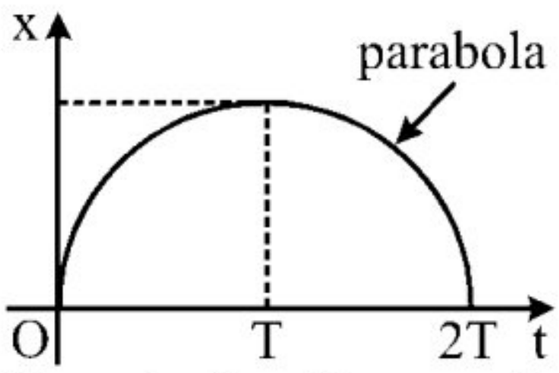
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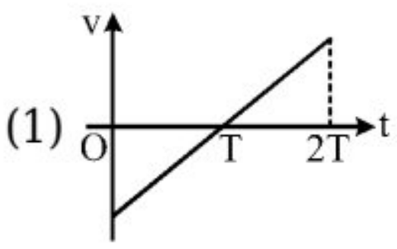
(3) $\frac{\alpha+\beta}{\alpha\beta}t$

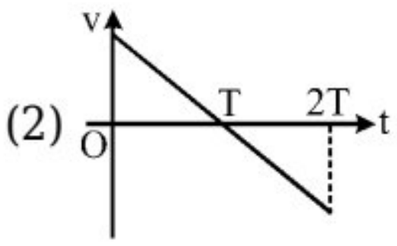
(4) $\frac{\alpha\beta}{\alpha+\beta}t$

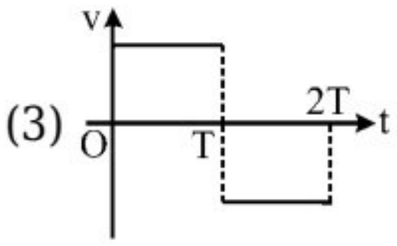
30) The x – t graph of a particle moving along a straight line is shown in figure.

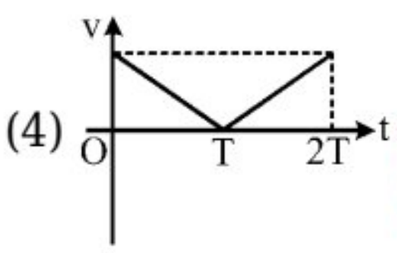


The velocity – time graph of the particle is correctly shows by :

(1) 

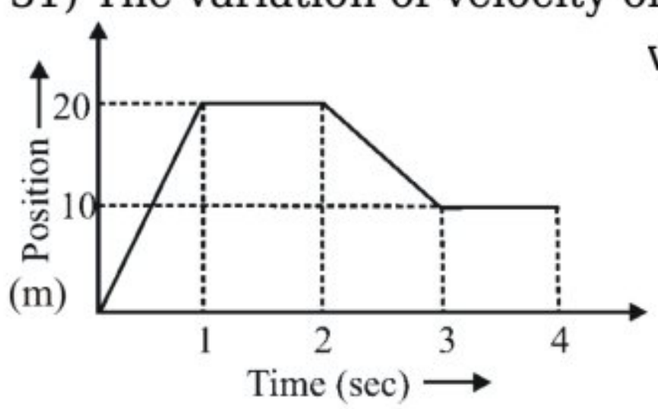
(2) 

(3) 

(4) 

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31) The variation of velocity of a particle moving along straight line is shown in figure. The average velocity of the body in 4 seconds is :-



(1) zero

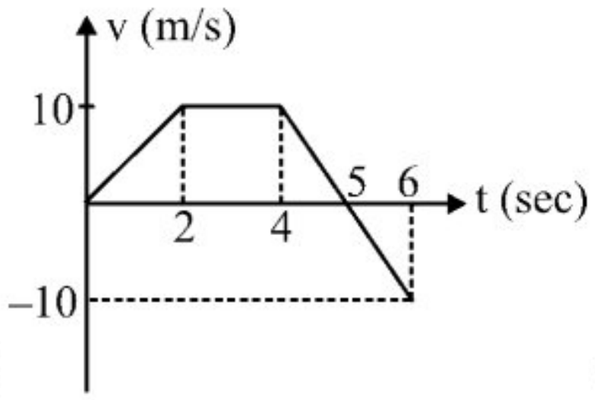
(2) 2.5 m/sec

(3) 10 m/sec

(4) 13.75 m/sec

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32) Calculate the average velocity for complete journey for the following graph?

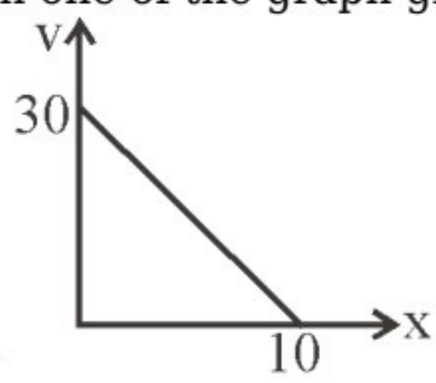
(1) 40 m/s

(2) 35 m/s

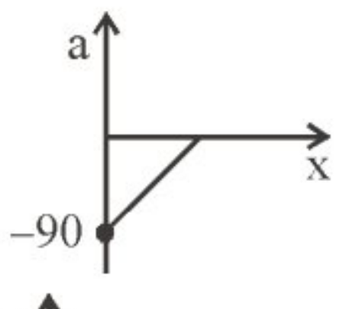
(3) 30 m/s

(4) 5 m/s

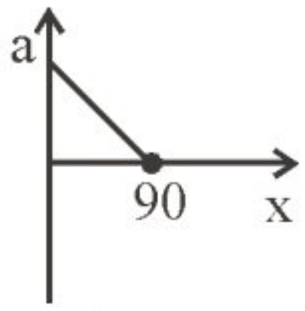
33) The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement :-



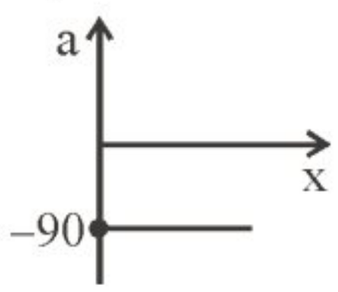
(1)



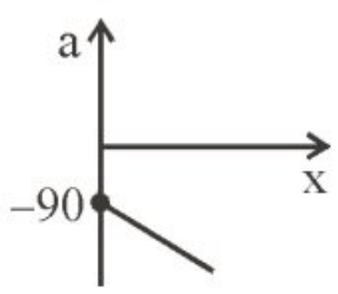
(2)



(3)



(4)



34) Drops of water falls from the roof of a building 18m. high at regular intervals of time. When the first drop reaches the ground, at the same instant fourth drop starts to fall. What are the distances of the second and third drops from the roof :

(1) 6 m and 2 m

(2) 6 m and 3 m

(3) 8 m and 2 m

(4) 4 m and 2 m

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35) A body is dropped from a height h above the ground. The ratio of the distances travelled in first second, second second, third second.....will be:-

(1) 1 : 3 : 5.....

(2) 1 : 4 : 9....

(3) 1 : 2 : 3.....

(4) 1 : 2 : 4....

36) A ball is thrown with velocity 600 m/s vertically upwards calculate distance travelled in last 2 sec of its upward motion :-

(1) 20 m

(2) 30 m

(3) 10 m

(4) 25 m

37) A particle is thrown vertical upwards, it crosses a point P in 4 sec and further move upward and take 8 sec more to reach the ground find height of P.

(1) 180 m

(2) 160 m

(3) 100 m

(4) 50 m

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38) Match the column:

Column-I	**Column-II**
(a) Freezing point of water	(p) 0 K
(b) Boiling point of water	(q) 0°C
(c) Maximum density of water	(r) 100°C
(d) Absolute zero	(s) 4°C

(1) (a)–(p), (b)–(r), (c)–(s), (d)–(r)

(2) (a)–(q), (b)–(r), (c)–(s), (d)–(p)

(3) (a)–(p), (b)–(s), (c)–(r), (d)–(q)

(4) (a)–(r), (b)–(q), (c)–(p), (d)–(s)

39) A solid material is supplied with heat at constant rate and the temperature of the material changes as shown below. From the graph, the false conclusion drawn is:

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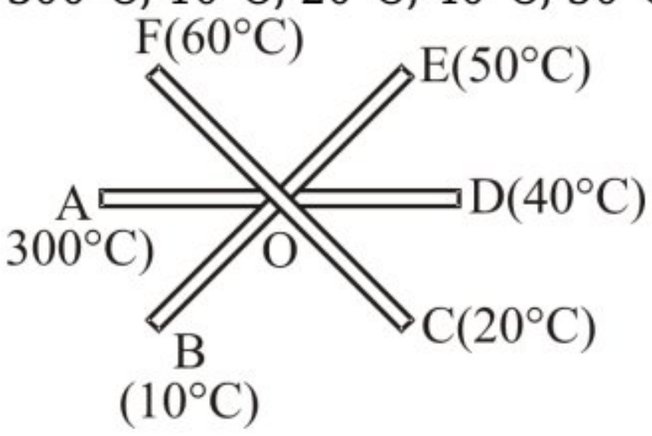
(1) AB and CD of the graph represent phase changes

(2) AB represent the change of state from solid to liquid

(3) latent heat of fusion is twice the latent heat of vaporization

(4) CD represents the change of state from liquid to vapour

40) Six identical rods are joined together as shown in figure temp of points A, B, C, D, E & F are-300°C, 10°C, 20°C, 40°C, 50°C, 60°C respectively find temp of Junction O.



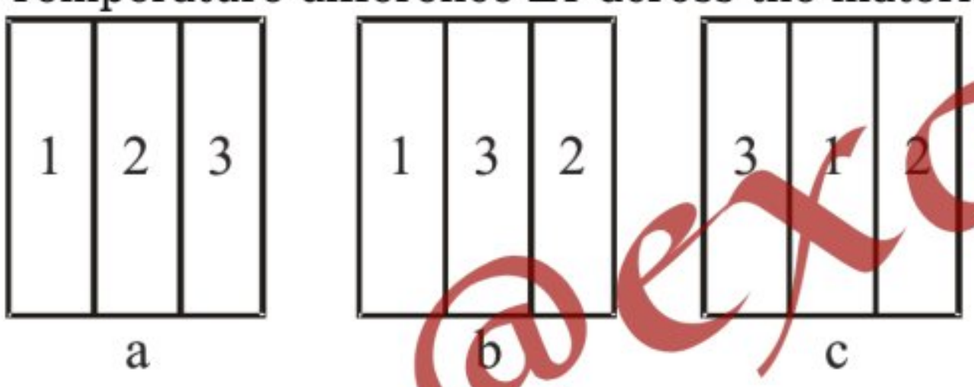
(1) 60°C

(2) 70°C

(3) 80°C

(4) None of these

41) Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is 20°C higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases :



(1) $\Delta T_a > \Delta T_b > \Delta T_c$

(2) $\Delta T_a = \Delta T_b = \Delta T_c$

(3) $\Delta T_a = \Delta T_b > \Delta T_c$

(4) $\Delta T_a = \Delta T_b < \Delta T_c$

42) **Assertion :** An object may have varying speed without having varying velocity.
Reason : If the velocity is zero at an instant, the acceleration should also be zero at that instant.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

43) A car moving with constant acceleration covered the distance between two points 60.0 m apart in 6.00 s. Its speed as it passes the second point was 15.0 m/s. What is the speed at the first point ?

(1) 10 m/s

(2) 5 m/s

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(3) 16 m/s

(4) 20 m/s

44)

A ball is thrown upwards from a planet such that distance travelled in the 4th and 5th second are equal. If the maximum height attained by the ball is 32m, choose the correct statement(s).
(A) Acceleration due to gravity of the planet is 2 m/s^2.
(B) Acceleration due to gravity of the planet is 4 m/s^2.
(C) Distance travelled by the ball in 4th second is 2m.
(D) Time taken for the ball to strike ground again is 10 s.

(1) A, C

(2) B, C

(3) B, C, D

(4) None of these

45) A body is dropped from a balloon moving up with a velocity of 4 m s^{-1} when the balloon is at a height of 120.5 m from the ground; the height of the body after five seconds from the ground is :-

(1) 8 m

(2) 12 m

(3) 18 m

(4) 24 m

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CHEMISTRY

1)

Assertion :- Ion-induced dipole interaction is stronger than dipole-induced dipole interaction.

Reason :- Accumulated charges on the molecule in case of ion-induced dipole is more compared to that in case of dipole interaction.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

2)

Assertion :- Molecular oxygen is attracted by magnetic field.

Reason :- Molecular oxygen contains 2 unpaired electrons which occupy two different p-molecular orbitals

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

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(4) Both Assertion & Reason are False.

3)

Which of the following overlaps is incorrect (assuming Z-axis as inter nuclear axis) ?
(i) $2P_y + 2p_y \rightarrow \pi$ Bond formation
(ii) $2P_x + 2p_x \rightarrow \pi$ Bond formation
(iii) $3d_{xy} + 3d_{xy} \rightarrow \pi$ Bond formation
(iv) $2s + 2p_y \rightarrow \pi$ Bond formation
(v) $3d_{xy} + 3d_{xy} \rightarrow \delta$ Bond formation

(vi) $2P_z + 2P_z \rightarrow \sigma$ Bond formation

(1) (i), (ii), (iii)
(2) (iii), (vi)
(3) (ii), (v)
(4) (iii), (iv)

4) Select reaction in which smallest change is observed in the % p-character.

(1) $NH_3 + HCl \longrightarrow NH_4Cl$
(2) $BF_3 + F^- \longrightarrow BF_4^-$
(3) $C_2H_6 \xrightarrow{h\nu} 2\dot{C}H_3$
(4) $BeCl_2 + Cl^- \longrightarrow BeCl_3^-$

5) Which among the following species is not perfectly planar ?

(1) $\dot{C}H_3$
(2) $\overset{\ominus}{C}HF_2$
(3) $^+CF_3$
(4) NO_2

6)

The species having no pπ-pπ bond but has bond order equal to that of O_2 :

(1) ClO_3^-
(2) PO_4^{3-}
(3) SO_4^{2-}
(4) XeO_3

7) In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed ?

(1) $N_2 \rightarrow N_2^{\oplus}$
(2) $C_2 \rightarrow C_2^{\oplus}$

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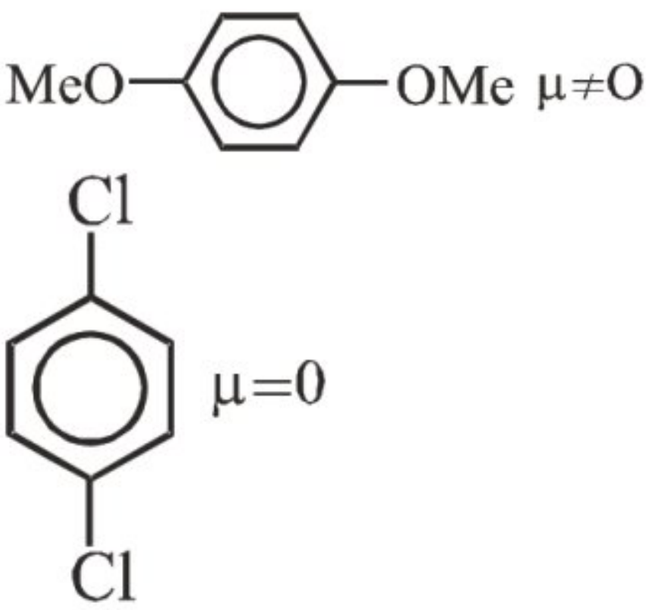
(3) $NO \rightarrow NO^{\oplus}$

(4) $O_2 \rightarrow O_2^{\oplus}$

8) Which is incorrect regarding dipole moment ?

(1) $CH_2Cl_2 > CHCl_3$

(2) $PCl_2Br_3 = PBr_2Cl_3$

(3) 

$\mu \neq O$

(4) $\mu = 0$

9) Which of the following has highest melting point :-

(1) NaCl

(2) LiCl

(3) LiBr

(4) NaI

10) Select wrong statement :

(1) transition metal ion is more polarising than s-Block ion of comparable size and charge

(2) solubility order is AgF > AgCl > AgBr > AgI

(3) LiCl is soluble in organic solvent

(4) None of the above is correct

11) Order of covalent character is not correct for :

(1) $NaF < Na_2O < Na_3N$

(2) $PbO < PbS$

(3) $CdCl_2 < SnCl_2$

(4) AgCl > NaCl

12) What is difference between bond angles in cationic species of PCl_5 and PBr_5 in solid state :

(1) 60°

(2) 109°28'

(3) 0°

(4) 90°

13) **Assertion** :- NCl_3 has pyramidal shape

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Reason :- In NCl_3 central atom is sp^3 hybridised.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

14) ***Assertion :-*** Bond length of P-Cl bonds in gaseous PCl_5 and Solid PCl_5 are not equal. ***Reason :-*** In solid state two PCl_5 molecules are associated as $[PCl_4^+]$ $[PCl_6^-]$.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

15) Cl–O bond length is minimum in :-

(1) ClO^-
(2) ClO_2^-
(3) ClO_3^-
(4) ClO_4^-

16) ***Assertion*** :- Solubility of alkali metal hydroxide increases on moving down the group.
Reason :- On moving down the group, hydration energy increases and lattice energy decreases.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) Assertion is True but the Reason is False
(4) Both Assertion & Reason are False

17) ***Assertion*** :- Na_2SO_4 is soluble in water while $BaSO_4$ is insoluble.
Reason :- Lattice energy of $BaSO_4$ exceeds its hydration energy.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

18)

Assertion :- Ionization energy of Ne is greater as compare to Ne^+.

Reason :- Ne has p^6 (full filled electronic configuration) but Ne^+ has p^4 configuration of outermost subshell.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

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(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

19)

Assertion :- Be and Mg have negative value of electron affinity, because.

Reason :- 2s and 3s orbitals are filled to their capacity and new coming e^- enters into new p subshell.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

20)

Assertion :- Electronegativity of an element is not always constant value.

Reason :- Electronegativity depends upon oxidation state and hybridisation state of atom.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

21) Atomic radius of Be, O, C, N, B are respecitvely (in pm) :

(1) 74, 84, 111, 77, 66

(2) 111, 66, 77, 74, 88

(3) 111, 88, 77, 74, 66

(4) 111, 66, 88, 77, 74

22)

Ionisation energies of element A are given below KJ/mol. :

IE_1	IE_2	IE_3
120	430	540

If A react with different elements, which compounds of A is not possible ?

(1) AF

(2) A_2O

(3) A_3N

(4) A_3N_2

23) The following acids have been arranged in the order of decreasing acid strength. Identify the correct order :-

ClOH (I) BrOH (II) IOH (III)

(1) I > II > III

(2) II > I > III

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(3) III > II > I

(4) I > III > II

24) In which of the following arrangements, the order is according to the property indicated against it ?

I	I < Br < F < Cl	Increasing electron gain enthalpy
II	Li < Na < K < Rb	Increasing metallic radius
III	B < C < N < O	Increasing first ionisation energy
IV	$Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$	Increasing ionic size

(1) I, II and III

(2) I and III only

(3) I, II and IV

(4) I and II only

25) Amphoteric behaviour is shown by the oxides of:-

(1) Al and Ca

(2) Pb and Ba

(3) Cr and Mg

(4) Sn and Zn

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26) If x, y and z are electronegativity, ionisation potential and electron-affinity respectively. Then the electron affinity (z) in the terms of electronegativity (x) and ionisation potential (y) will be :-

(1) $z = \frac{x+y}{2}$

(2) $z = \frac{x-y}{2}$

(3) $z = \frac{x^2-y^2}{2}$

(4) z = 2x – y

27) Which of the following is incorrect order :-

(1) Atomic radius :- $F < O < F^{-} < O^{-2}$

(2) 2nd Ionisation energy :- C < N < F < O

(3) Electron affinity :- I < Br < F < Cl

(4) Z_{eff} :- $Al < Al^{+} < Al^{+3} < Al^{+2}$

28) Which process is an exothermic process ?

(1) $N^{\ominus}_{(g)} \rightarrow N_{(g)} + e^{-}$

(2) $N^{+}_{(g)} \rightarrow N^{2+}_{(g)} + e^{-}$

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(3) $N_{(g)} + e^- \rightarrow N^{\ominus}_{(g)}$

(4) $N_{(g)} \rightarrow N^+_{(g)} + e^-$

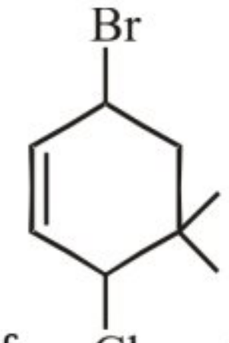
29) Which of the following oxides is not expected to react with sodium hydroxide :-

(1) BeO

(2) B_2O_3

(3) CaO

(4) SiO_2

30) IUPAC name of  is :-

(1) 3-Bromo-6-chloro-5,5-dimethylcyclohexene

(2) 6-Bromo-3-chloro-4,4-dimethylcyclohexene

(3) 3-Chloro-4,4-dimethyl-6-bromocyclohexene

(4) None of these

31) Which of the following statement is not true for homologous series ?

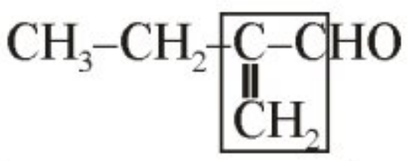
(1) Not having same molecular formula

(2) Not having same general formula

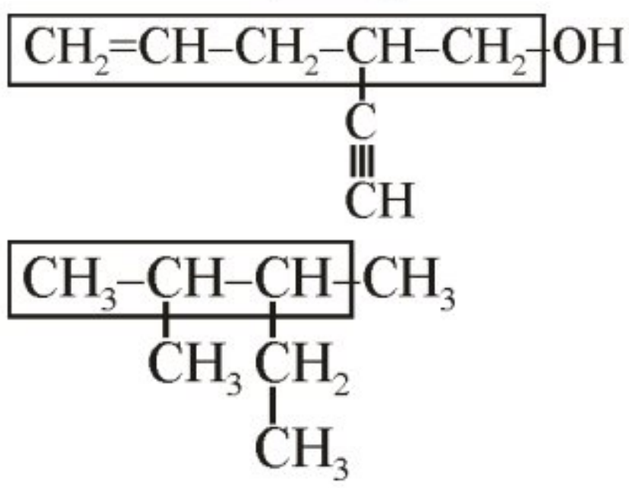
(3) Not having same chemical property

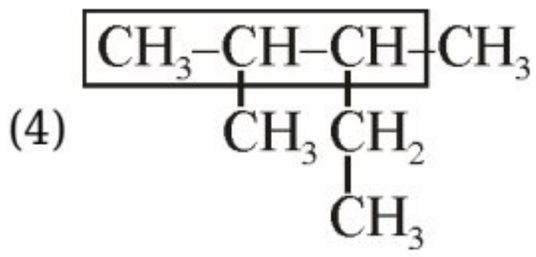
(4) 2 and 3

32) In which of the following parent chain is wrong ?

(1) 

(2) 

(3) 

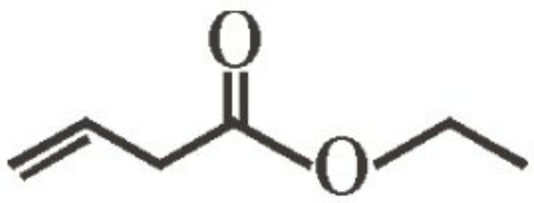
(4) 

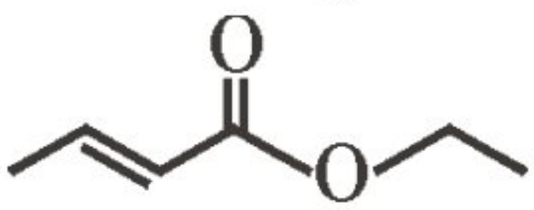
33) Correct structure of ethyl but-2-enoate is :-

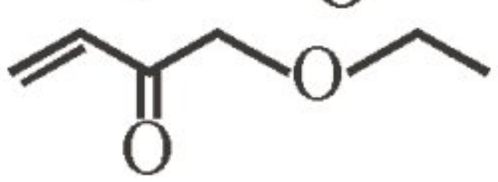
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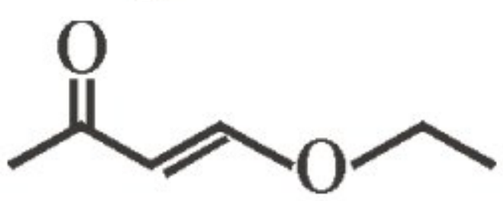
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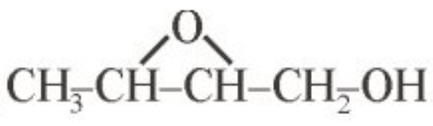
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(1) 

(2) 

(3) 

(4) 

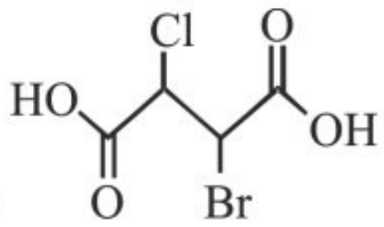
34) IUPAC name of  is :-

(1) 2, 3-Epoxy-4-butanol

(2) 2, 3-Epoxy-1-butanol

(3) 2, 3-Epoxy-1-hydroxy butane

(4) 1-Hydroxy-2,3-epoxy butane

35) **Assertion :-**  is named as 2-bromo-3-chlorobutane-1, 4-dioic acid and not 2-chloro-3-bromobutane-1, 4-dioic acid.

Reason :- If substituents are present at equivalent position than the substituent that comes first in alphabetical order gets less locant.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

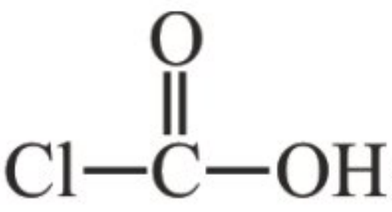
36) The derived name of isopentyl alcohol is:-

(1) Isobutyl carbinol

(2) Isopropyl carbinol

(3) Isopentyl carbinol

(4) Isopentanol

37) What is IUPAC of  is :-

(1) Chloro hydroxy ketone

(2) 1-chloro methanone

(3) 1-chloro methanoic acid

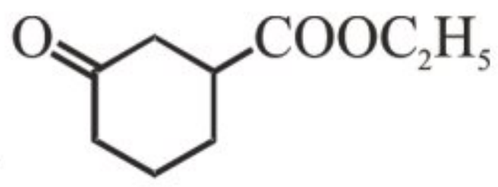
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(4) Hydroxy methanoyl chloride

38) The IUPAC of following will be:-



(1) Ethyl-3-oxo-cyclo hexane carboxylate

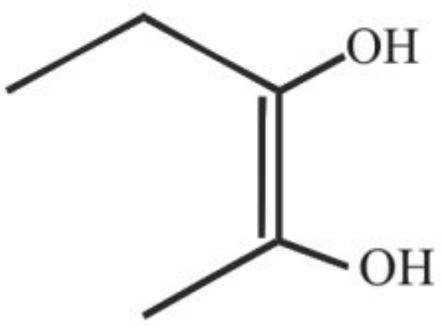
(2) Ethyl-3-keto cyclo hexanoate

(3) Ethyl-3-keto cyclo hexane ester

(4) Ethoxy cyclo hexanone

39)

Correct IUPAC name of the following is :-



(1) 1 - Methyl but - 1 - en -1, 2 - diol

(2) Pent - 3 - en - 3, 4 - diol

(3) Pent -2 - en - 2, 3 - diol

(4) 1 - ethyl - 2 - methyl ehtene - 1, 2 - diol

40) All the following IUPAC names are correct except

(1) 1–chloro–1–ethoxy propane

(2) 1–amino–1–ethoxy propane

(3) 1–ethoxy–2–propanol

(4) 1–ethoxy–1–propanamine

41) Which of the following is the correct IUPAC name of the structural formula given below :-



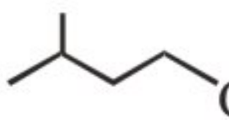
(1) 4 - Heptene - 2 - yn - 5 - amine

(2) 1 - Ethyl - 1 - Penten - 3 - yn - 1 - amine

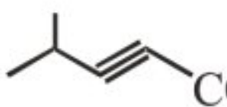
(3) 3 - Hepten - 5 - yn - 3 - amine

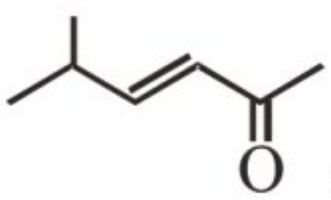
(4) 5 - Ethyl - 4 - Penten - 2 - yn - 5 - amine

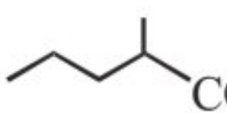
42) Indicate the wrongly named compound

(1)  CHO = 4-Methyl pentan-1-al

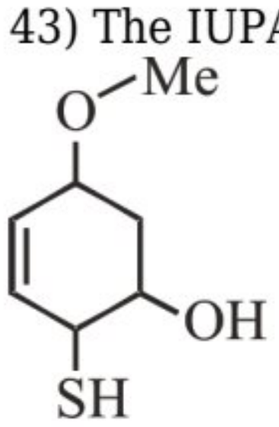
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(2)  = 4-Methyl pent-2-yn-l-oic acid

(3)  = Hex-3-en-5-one

(4)  = 2-Methylpentan-1-oic acid

43) The IUPAC name of given compound is :-



(1) 3-Methoxy-6-mercaptocyclohex-4-enol

(2) 6-Hydroxy-4-methoxycyclohex-2-ene-1-thiol

(3) 5-Methoxy-2-mercaptocyclohex-3-en-1-ol

(4) 2-Mercapto-5-methoxycyclohex-3-en-l-ol

44) Inter molecular H-bond :

(1) Decreases Volatility

(2) Increases melting point

(3) Increases viscosity

(4) All of these

45) The correct increasing order of adjacent bond angle among BF_3, PF_3 and CIF_3 :

(1) $BF_3 < PF_3 < ClF_3$

(2) $PF_3 < BF_3 < ClF_3$

(3) $ClF_3 < PF_3 < BF_3$

(4) $BF_3 = PF_3 = ClF_3$

BIOLOGY

1) Statement-I: Chrysophytes are microscopic and float passively in water currents.
Statement- II: The diatoms walls are embedded with slilca and are chief 'producers' in the oceans.

(1) Statement-I and Statement-II both are incorrect

(2) Statement-I and Statement-II both are correct

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

2) Analyse the following statements and identify the correct options given below :-
I. Two kindom system of classification did not distinguish between the eukaryotes and prokaryotes,

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unicellular and multicellular organism and algae and fungi.
II. The two kingdom classification used for a long time was found inadequate.

(1) I is true, but II is false

(2) I is false, II is true

(3) I and II are true

(4) I and II are false

3) Match List-I with List-II

	List-I		List-II
(A)	Ascomycetes	(i)	Albugo, Rhizopus
(B)	Basidiomycetes	(ii)	Alternaria, Trichoderma
(C)	Deuteromycetes	(iii)	Yeast, Neurospora
(D)	Phycomycetes	(iv)	Puccinia Agaricus

Choose the correct answer from the options given below:

(1) A–iv,B–iii, C–ii,D–i

(2) A–iii,B–iv, C–ii,D–i

(3) A–iii,B–iv, C–i,D–ii

(4) A–iv,B–iii, C–i,D–ii

4) According to R.H. Whittaker's classification system, the group of organism which is heterotrophic eukaryotic, multicellular organism lacking cell wall will be :-

(1) Monera

(2) Protista

(3) Plantae

(4) Animalia

5) Two kingdom system was mainly based on:-

(1) Cell wall

(2) Cell membrane

(3) Chloroplast

(4) Reserve food material

6) Water bloom is generally caused by :-

(1) Green algae

(2) Blue-green algae

(3) Bacteria

(4) Hydrilla

7) Bacteria that convert nitrate from nitrite in N_2 cycle is :

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(1) *Nitrobacter*

(2) *Nitrosomonas*

(3) *Azotobacter*

(4) *Rhizobium*

8) Which is common in Lichen and Mycorrhiza ?

(1) Algae

(2) Fungi

(3) Bacteria

(4) Protists

9) Which one of the following options gives the **correct** categorisation of examples with respect to their group to which they actually belong

	A	**B**	**C**
	Monera	**Protista**	**Fungi**
(1)	Cyanobacteria, Mycoplasma	Slime Moulds, *Euglena*	*Puccinia, Ustilago*
(2)	Cyanobacteria, Mycoplasma	*Gonyaulax, Euglena*	*Chlorella, Puccinia*
(3)	*Trichoderma, Colletotrichum*	*Gonyaulax, Euglena*	*Puccinia, Albugo*
(4)	BGA, *Penicillium*	Dinoflagellate, Chrysophytes	*Alternaria, Aspergillus*

(1) 1

(2) 2

(3) 3

(4) 4

10) A Gram negative bacteria will appear _____ after washing by alcohol.

(1) Red/Pink

(2) Deep blue/ purple

(3) Colourless

(4) Orange

11) Lichens are indicator of higher amount of which pollutant in air-

(1) SO_2

(2) CO_2

(3) CH_4

(4) NO_2

12) Identify the fungi on the basis of statements A and B and choose the correct option
A. Extensively used in biochemical and genetic research.
B. Used for making bread and beer.

(1) A - *Neurospora*, B - *Penicilliun*

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(2) A - *Penicillium*, B - Yeast

(3) A - *Neurospora*, B - Yeast

(4) A - *Trichoderma*, B - *Aspergillus*

13) What is incorrect for chemosynthetic autotrophic bacteria :-

(1) Oxidise various organic and inorganic substances

(2) Use the energy released through oxidation for their ATP production

(3) Play great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur

(4) They are most abundant in nature

14) Which one of the following is not a criteria of five kingdom system?

(1) Cell type

(2) Body organisation

(3) Reserve food materials

(4) Phylogeny

15) Consider the following statements (A-D)
(A) Heterocysts are important sites for nitrogen fixation
(B) Cell wall of diatoms is indestructible due to presence of calcium carbonate
(C) In fungi cell wall is made up of chitin
(D) Algae show isogamous, anisogamous and oogamous type of sexual reproduction
How many of the above statement (s) is/are not correct?

(1) One

(2) Two

(3) Three

(4) Four

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16) Dikaryophase/Dikaryon formation is a specific characteristic of :-

(1) all fungal classes

(2) phycomycetes and ascomycetes

(3) only basidiomycetes

(4) ascomycetes and basidiomycetes

17) In general, viruses that infect plants have

(1) ss DNA

(2) ss RNA

(3) ds DNA

(4) ds RNA

18) Find out the correct match from the following table:-

	Column I	**Column II**

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(i)	Fimbriae	Cell adhesion
(ii)	Capsule	Protein Synthesis
(iii)	Flagella	Help in genetic recombination

(1) (i) and (ii)

(2) (ii) and (iii)

(3) Only (iii)

(4) Only (i)

19) Red colour of red sea is due to :-

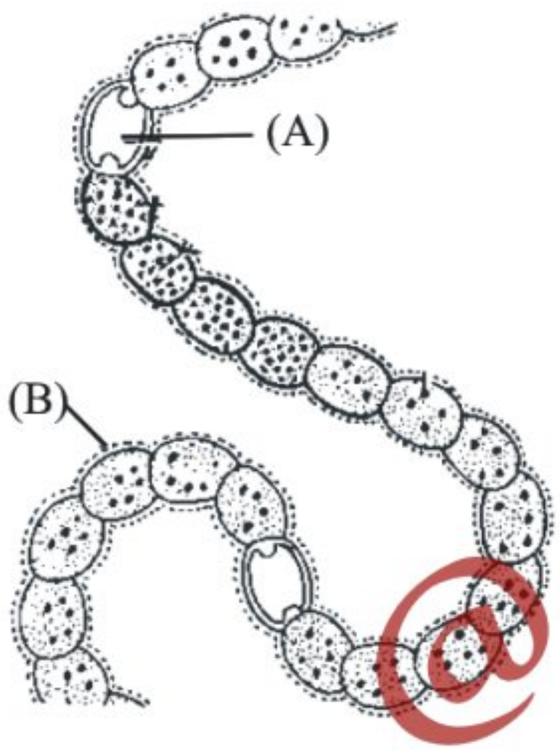
(1) *Gonyaulex*

(2) *Gymnodinium*

(3) *Trichodesmium erythrium*

(4) All of the above

20) Identify the given diagram and tick the correct answer.



Name	A	B
Nostoc	Heterocyst	Mucilagenous sheath
Chlamydo-monas	Heterocyst	Mucilagenous Sheath
Nostoc	Endospore	Outer cell membrane
Oscillatoria	Heterocyst	Chitin layer

(1) 1

(2) 2

(3) 3

(4) 4

21) Which one of the following structure is found in prokaryotes but lack in eukaryotes ?

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(1) Cell wall
(2) Ribosome
(3) Cilia
(4) Pili

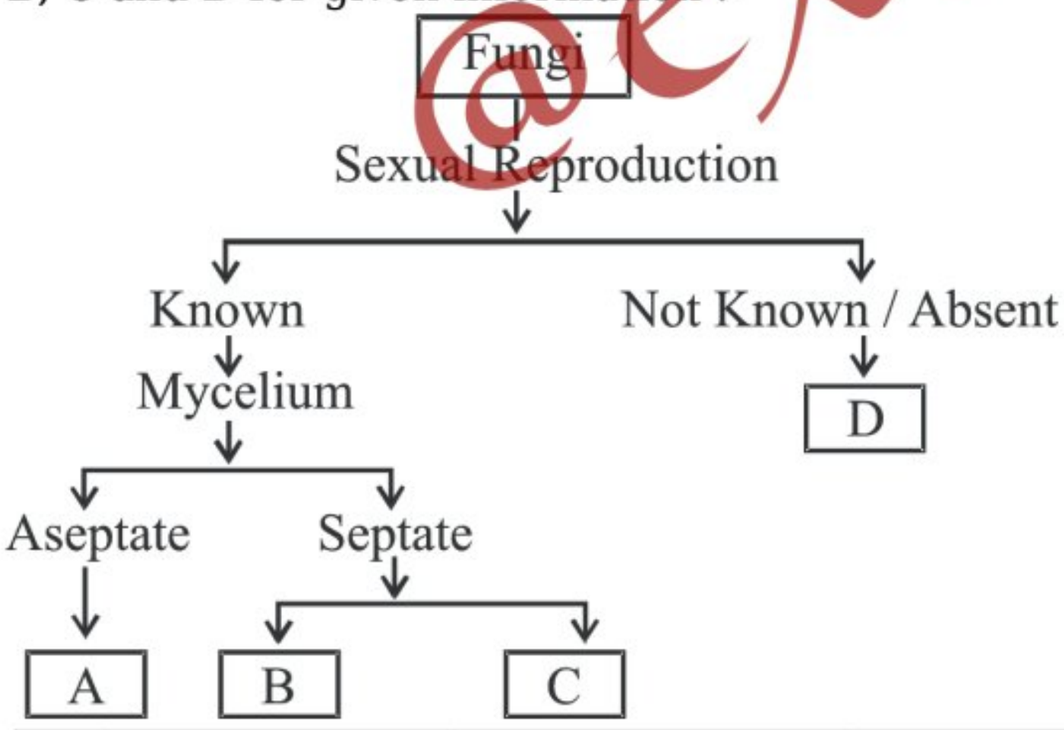
22) Potato spindle tuber disease is caused by :

(1) Bacteria
(2) Viroids
(3) Virus
(4) *Mycoplasma*

23) Read the following statements and choose the correct answer:
(A) Phycomycetes, mycelium is aseptate and coenocytic.
(B) The sexual cycle of fungi involves three steps: plasmogamy, karyogamy and meiosis.
(C) Ascospores are produced endogenously in ascus.
(D) Basidiospores are produced exogenously on the basidium
(E) Sexual reproduction in absent in deuteromycetes.

(1) Only C and D are incorrect
(2) Only A and E are incorrect
(3) Only A is incorrect
(4) All are correct

24) In given flow chart A, B, C, and D represent different classes of Kingdom Fungi. Identify the A, B, C and D for given information :-



	A	B	C	D
(1)	Phycomycetes	Deuteromycetes	Ascomycetes	Basidiomycetes
(2)	Ascomycetes	Phycomycetes	Basidiomycetes	Deuteromycetes
(3)	Deuteromycetes	Basidiomycetes	Ascomycetes	Phycomycetes
(4)	Phycomycetes	Ascomycetes	Basidiomycetes	Deuteromycetes

(1) 1
(2) 2

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(3) 3
(4) 4

25) Ascospores are formed in :-

(1) *Neurospora*
(2) *Penicilium*
(3) Yeast
(4) All of the above

26) The method of genetic recombination in bacterium in which a virus participates in :-

(1) Transformation
(2) Transduction
(3) Conjugation
(4) Translation

27) Most extensive metabolic diversity is found in :

(1) Fungi
(2) Algae
(3) Angiosperms
(4) Bacteria

28) In Whittaker's classification, eukaryotes are assigned

(1) Two of five kingdoms
(2) Three of five kingdoms
(3) Four of five kingdoms
(4) All the five kingdoms

29) Name the taxonomist who wrote "*Species plantarum*" and "*Systema Naturae*"

(1) Linnaeus
(2) Lamarck
(3) Darwin
(4) De Candolle

30) In a taxonomical hierarchy, class is incorporated between :-

(1) Phylum and order
(2) Division and order
(3) Order and genus
(4) More than one option is correct

31) In fungi which of the following is not produced during sexual reproduction ?

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(1) Oospore
(2) Zoospore
(3) Ascospore
(4) Basidiospore

32) Which is the outermost layer of cell envelop in eubacteria ?

(1) Cell wall
(2) Cell membrane
(3) Glycocalyx
(4) Middle lamella

33) Family solanaceae and convolvulaceae beongs to order :-

(1) *Poales*
(2) *Polymoniales*
(3) *Diptera*
(4) Dicot

34) Pigment phycocyanin and phycoerythrin are found in

(1) Bacillariophyceae
(2) Archaebacteria
(3) Eubacteria
(4) Cyanobacteria

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35) Which produces ascospores and basidiospores respectively ?

(1) Yeast and *Neurospora*
(2) *Neurospora* and *Agaricus*
(3) *Agaricus* and *Ustilago*
(4) *Agaricus* and Yeast

36) In angiosperms, floral features are commonly used for identification, why?

(1) Flower can be preserved easily
(2) Flower are attractive to work with
(3) Flower have aroma and different colour
(4) Reproductive parts are more conservative

37) In six kingdom classification_____organisms are placed in_____kingdoms.

(1) Eukaryotic, three
(2) Prokaryotic. three
(3) Eukaryotic, five
(4) Prokaryotic, two

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38) How many kingdoms in Linnaeus and Whittaker systems respectively show presence of chlorophyll 'a' in its members?

(1) Two; two

(2) Two; three

(3) One; three

(4) One; one

39) ***Assertion :-*** Slime moulds are protistan decomposers.
Reason :- Spores of slime moulds have cell walls.

(1) Assertion and Reason are true and the Reason is a correct explanation of the Assertion

(2) Assertion and Reason are true and the Reason is not a correct explanation of the Assertion

(3) Assertion is true but the Reason is false

(4) Both Assertion and Reason are false

40) The term "Systematics" was given by-

(1) Linnaeus

(2) Gaspard Bauhin

(3) Ernst Mayer

(4) Aristotle

41) Which of the following is true about gonyaulax ?

(1) Release toxin

(2) Slow multiplication causing red tide

(3) Heterotropic

(4) All of these

42) ``Contangium Vivum Fluidum'' term associated with .....

(1) Slime mould by Bejirinks

(2) Virus by Beijerineck

(3) Bacteria by Bejirinks

(4) Mycoplasma by Stanley Miller

43) Prions are mainly made of :-

(1) Protein

(2) DNA

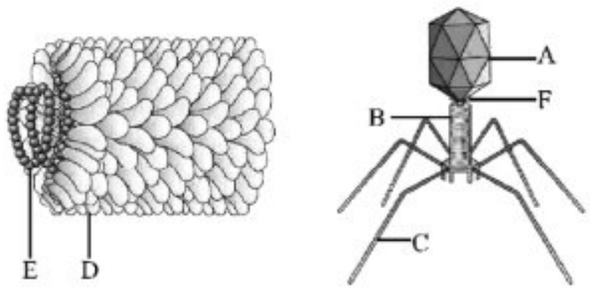
(3) RNA

(4) Both (2) & (3)

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44) Identify (B), (D) and (F) in these figures

(1)	B=Collar	D=RNA	F=Sheath
(2)	B=Capsid	D=Sheath	F=DNA
(3)	B=Sheath	D=Capsid	F=Collar
(4)	B=sheath	D=Capsid	F=RNA

(1) 1
(2) 2
(3) 3
(4) 4

45) Fill in the blanks by choosing suitable option for A,B,C & D respectively
" All taxonomic categories together constitute taxonomic hierarchy. Each category represents a ___A ___ and termed as __B __. Taxonomic categories are _____C _____ and not merely _____ D _____."

	(A)	**(B)**	**(C)**	**(D)**
(1)	Rank	Taxon	Taxonomic hierarchy	Taxonomic groups
(2)	Taxon	Rank	Taxonomic hierarchy	Taxonomic groups
(3)	Rank	Taxon	Biological Entities	Morpholigical aggregates
(4)	Taxon	Rank	Morpholigical aggregates	Biological Entities

(1) 1
(2) 2
(3) 3
(4) 4

46) (I) Triploblastic
(II) Unsegmented
(III) Acoelomate
(IV) Bilateral symmetry
These above description belongs to which phylum?

(1) Annelida
(2) Ctenophora
(3) Cnidarian
(4) Platyhelminthes

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47) Presence of external rows of ciliated comb plates is characterstic feature of -

(1) Cnidarians
(2) Flatworms
(3) Ctenophores
(4) Roundworms

48) Which of the following animals do not belong to class Mammalia?

(A) 

(B) 

(C) 

(D) 

(E) 

(1) B and E
(2) A and C
(3) E and C
(4) D and E

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49) Organ system level of organization is not found in:

(1) Annelida
(2) Arthropoda
(3) Mollusca
(4) Porifera

50) Which class of vertebrates bear jaw but does not bear limbs?

(1) Cyclostomata
(2) Chondricthyes
(3) Amphibia
(4) Reptilia

51) Match each item in Column I with one in Column II and select the best match from the codes given

Column -I		Column -II	
A	Comb plates	P	Cnidaria

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B	Metagenesis	Q	Mollusca
C	Canal system	R	Sponges
D	Radula	S	Ctenophora

(1) A-S, B-P, C-R, D-Q

(2) A-P, B-S, C-R, D-Q

(3) A-P, B-S, C-Q, D-R

(4) A-S, B-P, C-Q, D-R

52) In chordates

I. Heart is dorsal
II. Central nervous system is ventral
III. A post anal tail is present

(1) Only I & II are correct.

(2) Only III is correct.

(3) Only II is correct.

(4) Only I & III are correct.

53) Which of these possess a 4 chambered heart ?

(1) Fishes

(2) Frog

(3) Most reptiles

(4) Pigeon

54)

(P) Open circulation: arthropods (sinuses).
(Q) Closed circulation: annelids/chordates.
(R) Fish: 2-chambered heart, single circulation.
(S) Amphibians/reptiles: 3-chambered heart, incomplete circulation.
(T) Birds/mammals: 4-chambered heart, complete double circulation.
Mark the correct option based on the information given above.

(1) All are true

(2) Arthropods : Closed system

(3) Mammals : 3-chambered heart

(4) Fish : double circulation

55)

Match the following list of animals with their level of organisation.

Division of Labour		**Animal**	
A.	Organ level	1.	*Pheretima*
B.	Cellular level	2.	*Fasciola*

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C.	Tissue level	3.	*Spongilla*
D.	Organ system level	4.	*Obelia*

Choose the correct match showing division of labour with animal example.

(1) A–2, B–3, C–4, D–1

(2) A–2, B–4, C–3, D–1

(3) A–4, B–1, C–2, D–3

(4) A–1, B–4, C–3, D–2

56)

Select the correct difference w.r.t. the notochord in the following:

	Urochordata	**Cephalochordata**
(1)	Present only in larval tail	Extend from head to tail throughout life
(2)	Present only in adult	Present only in larval tail
(3)	Persistent throughouttheir life	Present only in adult
(4)	Extend from head to tail throughout life	Present only in larval tail

(1) 1

(2) 2

(3) 3

(4) 4

57) Which of the following groups contain only amphibians?

(1) *Bufo, Hyla, Ichthyophis, Salamandra, Rana*

(2) *Bufo, Rana, Crocodilus, Hyla, Salamandra*

(3) *Salamandra, Rana, Hemidactylus, Hyla, Bufo*

(4) *Hyla, Bufo, Naja, Ichthyophis, Salamandra*

58) Choose the animals that belong to super class tetrapoda:

(1) *Anopheles, Nereis, Balanoglossus, Ancylostoma*

(2) *Neophron, Ornithorhynchus, Hemidactylus, Ichthyophis*

(3) *Scoliodon, Bangarus, Petromyzon, Branchiostoma*

(4) *Pleurobrachia, Meandrina, Balanoglossus, Felis*

59) **Statement I.** Sponges are diploblastic animals.
Statement II. Sponges show cellular level of organisation.

(1) Only I correct

(2) Only II correct

(3) Both correct

(4) Both incorrect

60) **Assertion (A):** Arthropoda is the largest phylum.

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Reason (R): They have balancing organ statocyst.

(1) Both A & R true, R explains A

(2) Both true, R does not explain A

(3) A true, R false

(4) A false, R true; 5. Both false)

61) **Statement I.** Cnidarians have a complete digestive tract.
Statement II. Cnidarians show metagenesis.

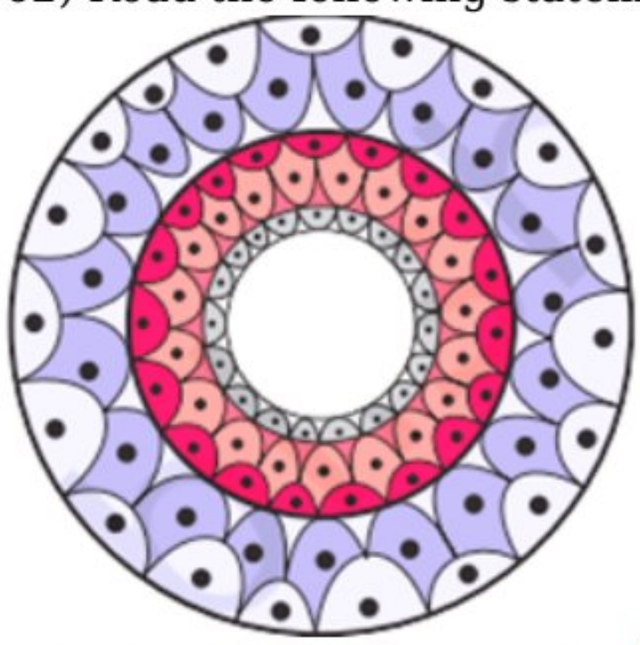
(1) Only 1 correct

(2) Only 2 correct

(3) Both correct

(4) Both incorrect

62) Read the following statements regarding the given diagrammatic sectional view-



I. Such a body cavity is called pseudocoelom.
II. Both Aschelminthes and Annelids have this type of coelom.

(1) Statement I is true but II is false.

(2) Statement I is false but II is true.

(3) Both statement are true.

(4) Both statements are false.

63) **Assertion (A):** *Ctenoplana* and *Pleurobrachia* are exclusively marine.
Reason (R): Ctenophores exhibit bioluminescence and lack cnidoblasts.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

64) Diploblastic and acoelomate animals are exemplified by :

(1) Arthropoda

(2) Coelenterata

(3) Platyhelminthes

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(4) Annelida

65) Metagenesis refers to

(1) Alternation between haploid and diploid generations.
(2) Alternation between sexual and asexual generations in the life cycle.
(3) Repeated asexual reproduction.
(4) Transformation of larva into adult.

66)
Choose the incorrect statements from the following :
A. Hemichordates have a rudimentary structure in the collar region.
B. Vertebrates have a ventral muscular heart.
C. Well developed muscular pharynx is present in Cnidarians.
D. The space between the mantle and shell is called mantle cavity.

(1) A and B
(2) B, C and D
(3) Only A
(4) C and D

67) Which of the following is a correct combination of characters related to the given animal?



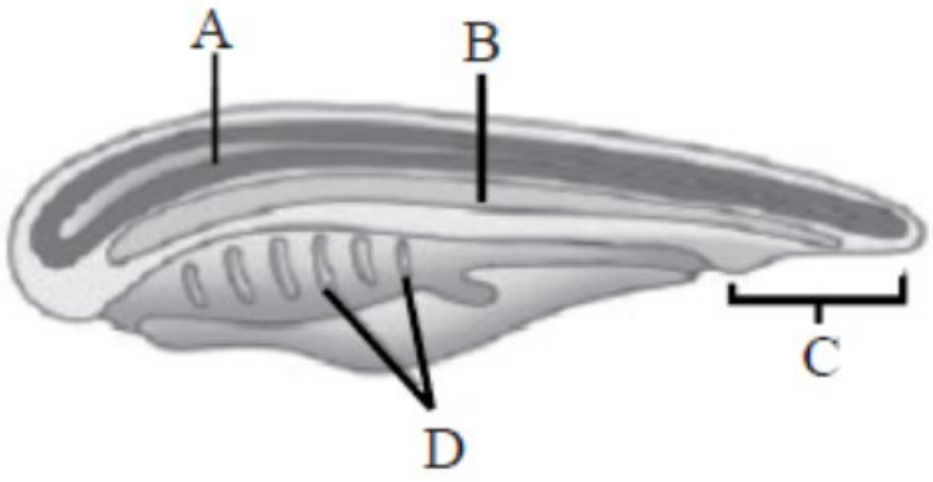
(1) Jaws present, paired fins absent, bony skeleton
(2) Jaws absent, paired fins absent, cartilaginous skeleton
(3) Jaws absent, paired fins present, cartilaginous skeleton
(4) Jaws present, paired fins present, cartilaginous skeleton

68) In which of the following animals, digestive tract has additional chambers like crop and gizzard?

(1) Bufo, Balaenoptera, Bangarus
(2) Catla, Columba, Crocodilus
(3) Pavo, Psittacula, Corvus
(4) Corvus, Columba, Chameleon

69) The given figure shows some characteristic features marked as chordates. Identify the correct labelling A,B,C and D.

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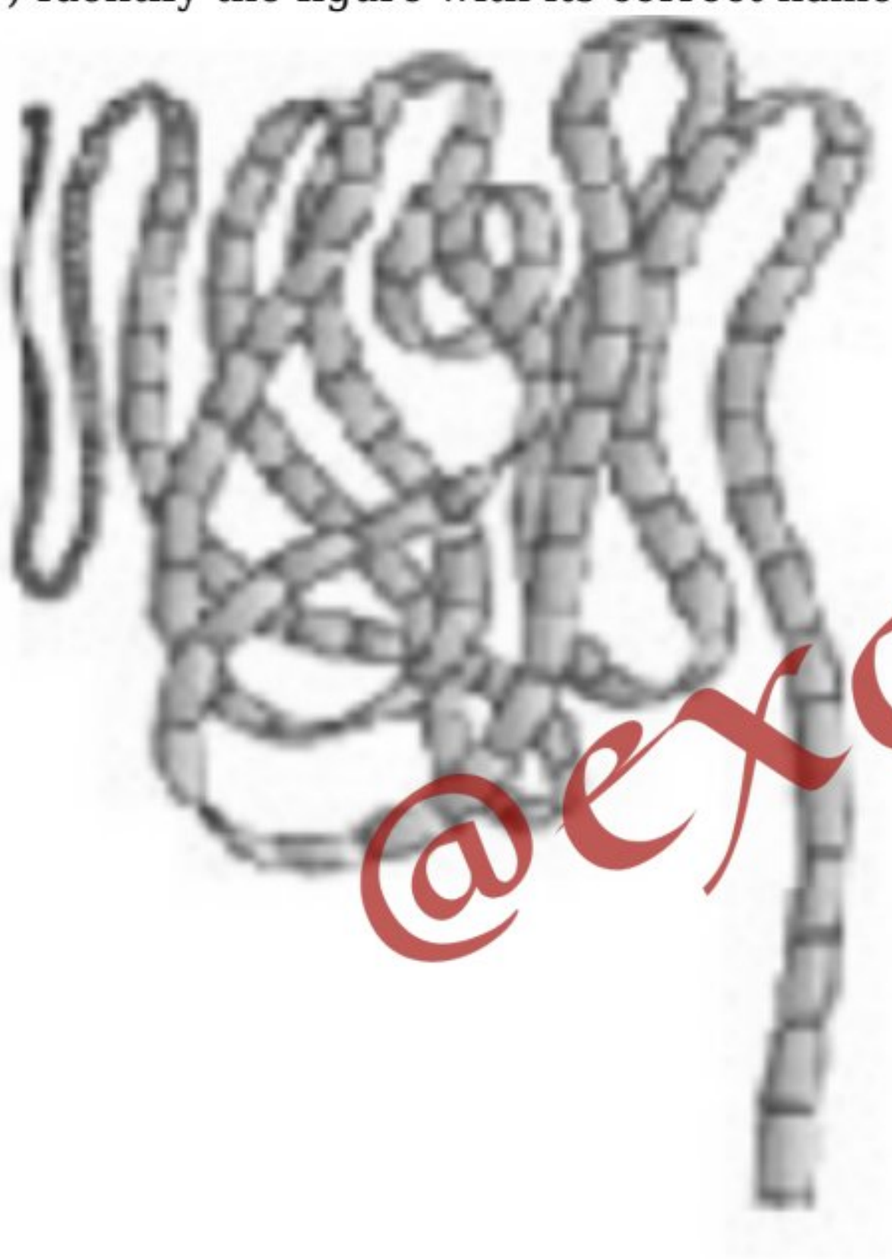
(1) A-Notochord; B-Post-anal part; C-Gill slits; D-Nerve cord

(2) A-Nerve cord; B-Notochord; C-Post-anal part; D-Gill slits

(3) A-Notochord; B-Nerve cord; C-Gill slits; D-Post-anal part

(4) A-Gill slits; B-Post-anal part; C-Nerve cord; DNotochord

70) Identify the figure with its correct name and phylum.



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(1) *Sycon* - Porifera

(2) *Aurelia* - Coelenterata

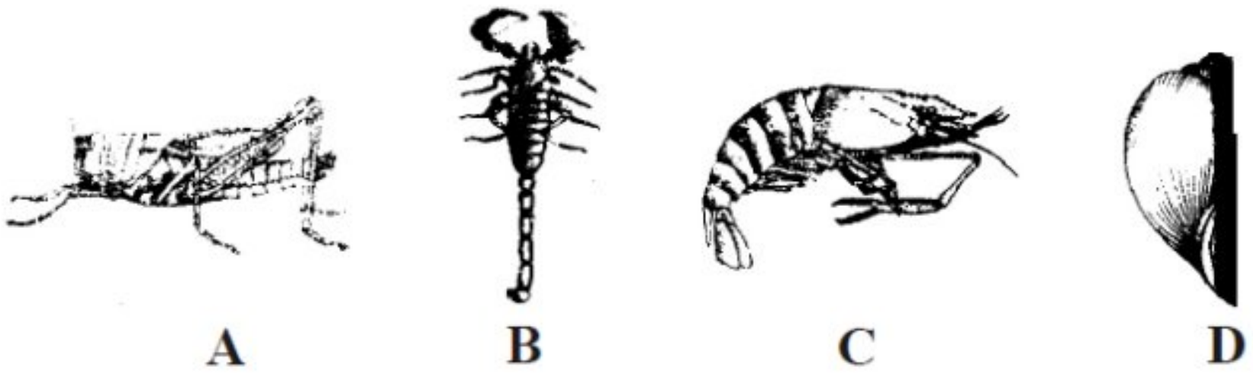
(3) *Pleurobrachia* - Ctenophora

(4) Tapeworm - Platyhelminthes

71) Refer the figures A, B, C and D given below. Which of the following options shows the correct name of the animals shown by the figures A, B, C and D ?

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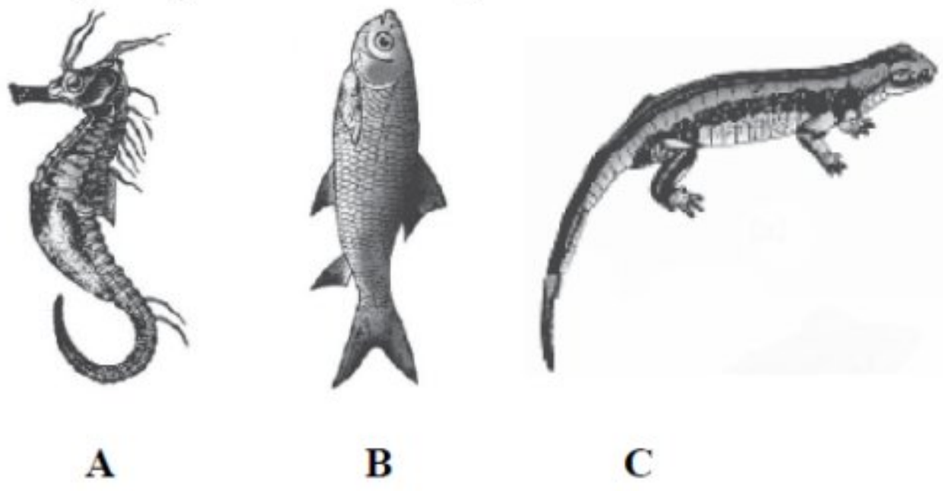
(1) A – Locust, B – Scorpion, C – Prawn, D – *Pila*

(2) A – Locust, B – Prawn, C – Scorpion, D – *Pila*

(3) A – Locust, B – Scorpion, C – Prawn, D – Snail

(4) A – Butterfly, B – Scorpion, C – Prawn, D – *Pila*

72) Refer the figures A, B and C and choose the correct option which shows animals that regulate buoyancy with the help of air bladder.



(1) A and B

(2) A and C

(3) B and C

(4) All of the above.

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73) Which of the following is not triploblastic?

(1) Flatworm

(2) Roundworm

(3) Hydra

(4) Earthworm

74) Which of the following is a connecting link between non-chordates and chordates?

(1) Balanoglossus

(2) Herdmania

(3) Amphioxus

(4) Petromyzon

75) The larvae of which phylum show bilateral symmetry while adults show radial symmetry?

(1) Cnidaria

(2) Mollusca

(3) Echinodermata

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(4) Porifera

76) Which structure in Hemichordata is a rudimentary outgrowth similar to a notochord?

(1) Proboscis gland
(2) Stomochord
(3) Post-anal tail
(4) Dorsal hollow nerve cord

77) Hemichordates are:

(1) Worm-like with proboscis, collar and trunk
(2) Spiny-skinned
(3) Soft-bodied with shell
(4) Jointed-legged

78) Moist skin without scales, three-chambered heart, and dual mode of life are characteristics of:

(1) Reptilia
(2) Amphibia
(3) Aves
(4) Mammalia

79) Book lungs or book gills are respiratory organs in:

(1) Insects
(2) Arachnids and some crustaceans
(3) Annelids
(4) Molluscs

80) Primitive multicullular animals belongs to :-

(1) Porifera
(2) Coelenterata
(3) Ctenophora
(4) Protozoa

81) Placoid scales and cartilaginous endoskeleton are features of:

(1) Osteichthyes
(2) Chondrichthyes (Sharks, Rays)
(3) Amphibia
(4) Reptilia

82) The rasping feeding organ called radula is present in:

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(1) Arthropoda

(2) Mollusca

(3) Annelida

(4) Echinodermata

83) **Statement I:** Aurelia (Jellyfish) represents the Medusa stage.
Statement II: Hydra represents the Polyp stage.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement I is correct, statement II is incorrect.

(4) Statement I is incorrect, statement II is correct.

84) Which bird is flightless?

(1) Psittacula

(2) Struthio

(3) Columba

(4) Neophron

85) Organ-system level of organization first appeared in which phylum?

(1) Porifera

(2) Aschelminthes

(3) Annelida

(4) Arthropoda

86) Which of the following is a "living fossil"?

(1) Archaeopteryx

(2) Limulus

(3) Pavo

(4) Neophron

87) The "Poison sting" in Chondrichthyes is found in:

(1) Torpedo

(2) Trygon

(3) Scoliodon

(4) Carcharodon

88) The presence of a "sucking and circular mouth" without jaws is the hallmark of:

(1) Exocoetus

(2) Pristis

(3) Myxine

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(4) Clarias

89) Which of the following is a true fish:

(1) Silverfish
(2) Cutlefish
(3) Dog fish
(4) Starfish

90) Metameric segmentation (true segmentation with repetition of organs) is best seen in:

(1) Porifera
(2) Annelida
(3) Mollusca
(4) Echinodermata

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NOVA

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	4	3	3	2	3	2	2	4	3	1	2	4	3	3	2	3	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	3	4	3	1	2	4	4	2	2	4	1	3	1	1	2	2	3	3
Q.	41	42	43	44	45															
A.	2	4	2	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	4	1	2	4	3	2	1	4	3	3	2	1	4	3	1	4	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	1	3	4	4	4	1	3	2	4	4	2	2	1	1	3	1	3	2
Q.	86	87	88	89	90															
A.	3	3	4	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	2	4	1	2	1	2	1	3	1	3	4	3	1	4	2	4	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	4	4	4	2	4	3	1	4	2	3	2	4	2	4	4	3	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	1	3	3	4	3	3	4	2	1	2	4	1	1	1	1	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	2	2	2	4	2	3	2	4	1	1	3	1	3	2	1	2	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	1	2	2	2	2	3	3	2										

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SOLUTIONS

PHYSICS

1) $\frac{x - MP}{BP - MP} = \frac{°C}{100}$

$\Rightarrow \frac{x - 39}{239 - 39} = \frac{39}{100}$

$\Rightarrow x = 117\ °W$

2) NCERT XI Pg # 300

3)

Real expansion of liquid in vessel A
= Real expansion of the liquid in vessel B

$\Rightarrow \gamma_{r1} = \gamma_{r2}$

$\Rightarrow \gamma_1 + 3\alpha = \gamma_2 + 3\alpha_B$

$\Rightarrow \alpha_B = \frac{\gamma_1 - \gamma_2}{3} + \alpha$

4) $\gamma = \alpha_x + \alpha_y + \alpha_Z$

$\gamma = \alpha_1 + \alpha_2 + \alpha_2$

$= \alpha_1 + 2\alpha_2$

5)

Question Explanation : Asking about the temperature at which the pendulum clock will give correct time.

Concept : This question is based on Effect of temperature on pendulum clock.

Formula: $T = 2\pi\sqrt{\frac{\ell}{g}}$

Solution :

From the Relation between time period of pendulum & time taken.

Time Period (t) $\propto \sqrt{\ell}$

$\frac{\Delta t}{t_0} = \frac{1}{2}\left(\frac{\Delta \ell}{\ell}\right)$

 From thermal expiation $\frac{\Delta \ell}{\ell} = \propto(\Delta T)$

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$$\frac{\Delta t}{t_0} = \frac{1}{2} \propto (\Delta T)$$

Since the clock is fast at 15°C (T_1), the correct temperature must be higher than 15°C to expand the pendulum and slow it down.

Since the clock is slow at 30°C (T_2), the correct temperature must be lower than 30°C to contract the pendulum and speed it up.

Let, T is the temperature for correct time. ($T_1 < T < T_2$)

$$\frac{(\Delta t)_1}{(\Delta t)_2} = \frac{(T - T_1)}{(T_2 - T)}$$

$$\frac{5}{10} = \frac{T - 15}{30 - T}$$

$30 - T = 2T - 30$

$3T = 60$

$T = 20°C$

Final Answer: option (3)

6)

Explain Question : Asking about the bending direction of the bimetallic strip, placed in the cold bath.

Concept : This question is based on Thermal expiation in Bimetallic strip.

Solution :

As $\alpha_A > \alpha_B$ so contraction in metal A is more than metal B.
that, means if A contracts more, So that strip will be bend towards A or lift side.

Final Answer: option (2)

7) **Explain :** We are asked to find the coefficient of linear expansion of a metal when the volume of a metal sphere increases by 0.15% when its temperature is raised by 24°C.
The relationship between volume change and temperature change for a solid object is linked to the coefficient of linear expansion (α), and we need to use this to calculate the required value.

Concept : Coefficient of linear expansion.

Formula The change in volume (ΔV) of a solid is related to the initial volume (V_0) and the coefficient of volumetric expansion (β) as follow.

$$\gamma = \frac{\Delta V}{V} \times \frac{1}{\Delta T}$$

Calculation :

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$$\gamma = \frac{0.15}{100} \times \frac{1}{24} \Rightarrow \alpha = \frac{\gamma}{3}$$

$$\alpha = \frac{0.15}{100} \times \frac{1}{24} \times \frac{1}{3} = 2.0 \times 10^{-5}/°C$$

Final Answer : Option (2)

8) **Question Explanation :** 20 g of aluminium with specific heat 0.2 cal/g°C is given. Find its water equivalent, i.e., the mass of water having the same heat capacity.

Concept: Calorimetry → Water Equivalent

Formula : Water equivalent = m × s

Solution: Given:

m = 20 g, s = 0.2 cal/g°C

Water equivalent = 20 × 0.2 = 4 g

Final answer: Option (2)

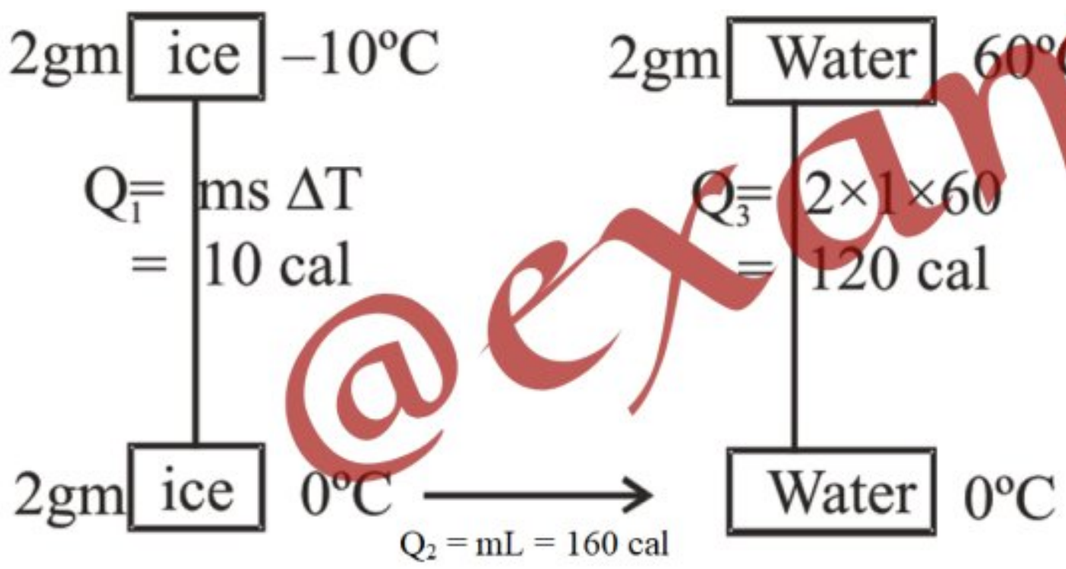
9) **Explain Question:**

Asking about of heat required to convert 2 gm ice at –10°C into water at 60°C.

Concept:

Phase change, temperature change.

Solution:



Q = 10 + 160 + 120 = 290 cal.

Final answer: Option (2)

10) **Explain Question:**

Asking about representation of phase change given in heat curve for given amount of a substance.

Concept:

Heat curve, latent heat of vapourisation and fusion.

Solution:

Phase conversion is isothermal process

Final answer: Option (4)

11)

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Question Explanation : We are asked to find the specific heat of a liquid when it is mixed with another liquid of known specific heat and temperatures, given equal masses and final temperature.

Concept : This question is based on principle of calorimetry mixing in same phase.

Solution : For equal masses, heat lost by hotter liquid = heat gained by cooler liquid :

Let c_2 be the unknown specific heat of the second liquid.

$mc_1(T_1 - T_f) = mc_2(T_f - T_2)$

$c_1(T_1 - T_f) = c_2 (T_f - T_2)$

Given : $c_1 = 0.8$ cal/g°C, $T_1 = 60°C$, $T_2 = 45°C$, $T_f = 53°C$

$0.8(60 - 53) = c_2(53 - 45)$

$0.8(7) = c_2(8)$

$5.6 = 8c_2$

$c_2 = \frac{5.6}{8} = 0.7$ cal/g°Cl

Final answer : Option (3)

12) $mgh = \frac{mL}{5} \Rightarrow h = \frac{L}{5g}$

13) $420\ J = ms\Delta T + m'L$

$m \times \frac{1}{2} \times 5 + 1 \times 80 = \frac{420}{4.2}$ cal

$m = 8$ g

14) **Question Explanation:** To find the effective thermal conductivity (K_{eff}) of a system of three identical rods joined end- to-end.

Given Data:

Three identical rods (same length L and area A).

Thermal conductivity of C = K.

Thermal conductivity of B = 3K (thrice of C).

Thermal conductivity of A = 6K (since B is half of A).

Concept:

Thermal Conductors in Series.

Solution:

Formula: $\frac{L_{tatal}}{K_{eff}\,A} = \frac{L_A}{K_A A} + \frac{L_B}{K_B A} + \frac{L_C}{K_C A}$

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Calculation: Total length = 3L.

$$\frac{3L}{K_{eff}} = \frac{L}{6K} + \frac{L}{3K} + \frac{L}{K}$$

$$\frac{3}{K_{eff}} = \frac{1}{6K} + \frac{2}{6K} + \frac{6}{6K}$$

$$\frac{3}{K_{eff}} = \frac{9}{6K} \Longrightarrow \frac{3}{K_{eff}} = \frac{3}{2K}$$

$$K_{eff} = 2K$$

Final Answer: Option (4)

15)

Correct Answer is Option (3).

16) $\frac{dQ}{dt} = -KA\frac{dT}{dx}$

$$dT \propto \frac{1}{A}\left[\frac{dQ}{dt}, K, dx = \text{Same}\right]$$

17)

$R_1 = 2R \rightarrow Q \rightarrow \text{time} = t$

$R_2 = \frac{R}{2} \rightarrow Q \rightarrow \text{time} = \frac{t}{4}$

18)

Explanation :- Ice forms on a lake at 0° C while the air temperature is –10°C. Given that forming a 1 cm layer takes 7 hours, we need to find the time required for the ice to grow from 1 cm to 2 cm.

Concept- :- Rate of ice formation using heat conduction and thickness relation.

Formula :- $t \propto d^2$

where t is the time and d is the thickness of ice.

Calculation :-

- Given that forming 1 cm of ice takes 7 hours, we use the relation $t \propto d^2$.
- Time required to grow from 0 cm to 1 cm.

$t_1 = 7 \text{ hours} \propto (1)^2$

- Time required to grow from 0 cm to 2 cm.

$t_2 \propto (2)^2 = 4 \times 7 = 28 \text{ hours}$

- Time taken to grow from 1 cm to 2 cm.

$t_1 \times t_2 = t_2 - t_1 = 28 - 7 = 21 \text{ hours}$

Final answer :- Option 3

21 hours

19)

Correct Answer is Option (4).

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20)

Correct Answer is Option (3).

21)

Explanation the Question:
A car covers 2/5 of the total distance with speed v_1 and the remaining 3/5 distance with speed v_2. find the average speed for the entire journey.
Concept:
This question is based on average speed
Formula:

$$\text{Average speed } (v_{avg}) = \frac{\text{Total Distance}}{\text{Total Time}}$$

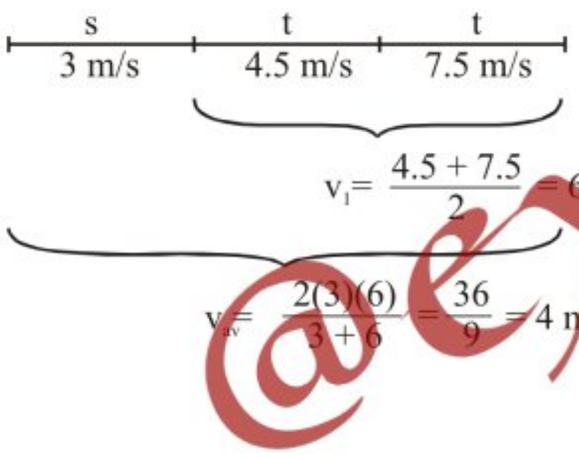
Solution:

$$V_{avg} = \frac{x}{\frac{\frac{2x}{5}}{V_1} + \frac{\frac{3x}{5}}{V_2}}$$

$$V_{avg} = \frac{5}{\frac{2}{V_1} + \frac{3}{V_2}} = \frac{5V_1V_2}{3V_1 + 2V_2}$$

Final Answer: Option (4)

22)



23)

Explaining the Question :- if $s = 3t^3 + 7t^2 + st + 8$ than acceleration of particle of t = 1 sec is
Concept:- This question is based on acceleration
Solution :- Given $s = 3t^3 + 7t^2 + st + 8$

$$v = \frac{ds}{dt} = \frac{3t^3 + 7t^2 + 5t + 8}{dt}$$

$$v = 9t^2 + 14t + 5$$

$$a = \frac{dv}{dt} = \frac{at^2 + 14t + 5}{dt}$$

$$a = 18t + 14$$

at t = 1 sec

$$a = 18 \times 1 + 14$$

$$a = 32 \text{ m/s}^2$$

Final Answer :- 3

24) $\frac{U}{g} = 6 \Rightarrow U = 60$ m/s

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$$S_{1st} = 60 - \frac{1}{2}(10)[2(1) - 1] = 55\,m$$

$$S_{7th} = 60 - \frac{1}{2}(10)[2(7) - 1] = -5\,m$$

$$\frac{S_{1st}}{S_{7th}} = \frac{55}{5} = \frac{11}{1}$$

25)

$x = \sqrt{V + 1}$

$x^2 = V + 1$

$V = x^2 - 1$

$a = V\frac{dV}{dx} = (x^2 - 1)(2x)$

at $x = 5$

$a = 24 \times 10 = 240$ unit

26)

$$a = \frac{dv}{dt} = 2(t-1) \Rightarrow \int_0^v dv = \int_0^5 (2t-2)dt$$

$$\Rightarrow v = [t^2 - 2t]_0^5 \Rightarrow 15\ m/s$$

27)

$\because x = 8t - 4t^2$

at $t = 0$, $x = 0$

$\because v = \frac{dx}{dt} = 8 - 8t$ if $v = 0$, $t = 1$

at $t = 1$, $x = 8 - 4 = 4m$

at $t = 2$, $x = 8(2) - 4(2)^2 = 0$

so, travelled distance $= |(4-0)| + |(0-4)|$

$= 4 + 4 = 8m$

28)

By $v^2 = u^2 + 2as$

$\left(\frac{u}{3}\right)^2 = u^2 - 2ax$

$2ax = u^2 - \frac{u^2}{9}$

$$\boxed{2ax = \frac{8u^2}{9}} \quad ...(1)$$

Similarly from starting

$v^2 = u^2 + 2ax$

$0 = u^2 - 2ax_2$

$$\boxed{2ax_2 = u^2} \quad ....(2)$$

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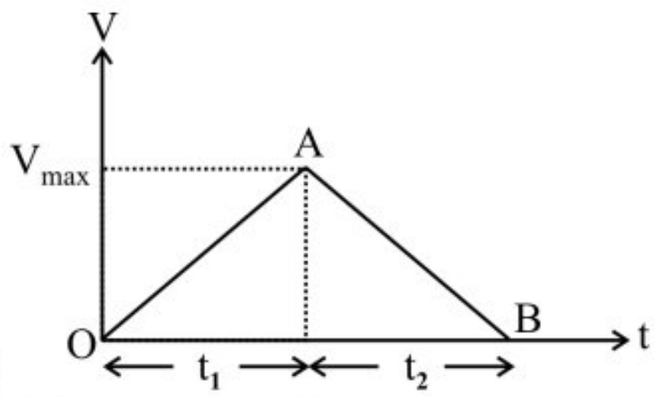
By (1) / (2)

$$\frac{x}{x_2} = \frac{8}{9}$$

$$\frac{24}{x_2} = \frac{8}{9}$$

$x_2 = 27$ cm

29)



$$\alpha = \frac{V_{max}}{t_1} \; ; \; \beta = \frac{V_{max}}{t_2}$$

$t = t_1 + t_2$

$$t = \frac{V_{max}}{\alpha} + \frac{V_{max}}{\beta}$$

$$v_{max} = \frac{\alpha\beta t}{\alpha + \beta}$$

30)

Correct Answer is Option (2).

31) $$Vavg = \frac{\text{change in position}}{\text{time}}$$

$$Vavg = \frac{10-0}{4} = 2.5 m/s$$

32) $$v_{avg} = \frac{\text{displacement}}{\text{time}} = \frac{\frac{1}{2}(5+2)\times 10 - \frac{1}{2}\times 1\times 10}{6}$$

$$v_{avg} = \frac{35-5}{6} = 5 m/s$$

33)

Correct Answer is Option (1).

34)

Correct Answer is Option (3).

35)

Correct Answer is Option (1).

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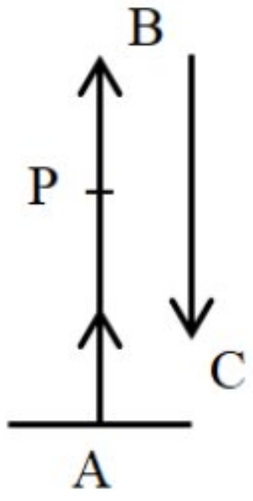
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36)

last 2 sec of a ball in upwards direction is same as downward motion in first 2sec so , a $S = \frac{1}{2} gt^2$

$S = \frac{1}{2} \times 10 \times 2^2$

20 m



37)

total time = 8 + 4 = 12

$t_{BC} = 6$

$t_{BP} = 2$

height of $P = \frac{1}{2} g(6)^2 - \frac{1}{2} g(2)^2$

$= 180 - 20 = 160$ m

38) Conceptual.

39)

Correct Answer is Option (3).

40)

Correct Answer is Option (3).

41) In all the three cases, temperature difference along the combination is same. Thus, heat current in all the three cases are same ⇒ temperature difference across any layer in all the three case is same.

42)

Correct Answer is Option (4).

43)

$\square s = \left(\frac{u+v}{2}\right) \times t, \quad \square 60 = \left(\frac{u+15}{2}\right) \times 6$

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$\therefore$ u = 5 m/s

44) $\because U = \sqrt{2gH} = \sqrt{32 \times 2g} = \sqrt{64g}$

$\because t = \frac{U}{g} = \frac{\sqrt{64g}}{g}$

$\therefore 4 = \sqrt{\frac{64}{g}} \quad \therefore g = 4\ m/s^2$

$\because S_{n^{th}} = U + \frac{a}{2}(2n-1)$

$\therefore S_{4^{th}} = \sqrt{64 \times 4} = \frac{4}{2}(2 \times 4 - 1)$

$= 16 \times 14 = 2$ m

45)

Correct Answer is Option (3).

CHEMISTRY

46)

Correct Answer is Option (1).

47)

Correct Answer is Option (1).

48)

Correct Answer is Option (4).

49)

Correct Answer is Option (1).

50)

Correct Answer is Option (2).

51)

Correct Answer is Option (4).

52)

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Correct Answer is Option (3).

53)

Correct Answer is Option (2).

54)

Correct Answer is Option (1).

55)

Correct Answer is Option (4).

56)

Correct Answer is Option (3).

57)

Correct Answer is Option (3).

58)

Correct Answer is Option (2).

59)

Correct Answer is Option (1).

60)

Correct Answer is Option (4).

61)

Correct Answer is Option (3).

62)

Correct Answer is Option (1).

63)

Correct Answer is Option (4).

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64)

Correct Answer is Option (1).

65)

Correct Answer is Option (1).

66)

Correct Answer is Option (2).

67)

Correct Answer is Option (4).

68)

Correct Answer is Option (1).

69)

Correct Answer is Option (3).

70)

Correct Answer is Option (4).

71)

Correct Answer is Option (4).

72)

Correct Answer is Option (4).

73)

Correct Answer is Option (1).

74)

Correct Answer is Option (3).

75)

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Correct Answer is Option (2).

76)

Correct Answer is Option (4).

77)

Correct Answer is Option (4).

78)

Correct Answer is Option (2).

79)

Correct Answer is Option (2).

80)

Correct Answer is Option (1).

81)

Correct Answer is Option (1).

82)

Correct Answer is Option (3).

83)

Correct Answer is Option (1).

84)

Correct Answer is Option (3).

85)

Correct Answer is Option (2).

86)

Correct Answer is Option (3).

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87)

Correct Answer is Option (3).

88)

Correct Answer is Option (4).

89)

Correct Answer is Option (4).

90)

Correct Answer is Option (3).

BIOLOGY

91)

Correct Answer is Option (2).

92) NCERT Pg.# 16,17

93)

Correct Answer is Option (2).

94) NCERT Pg.#17

95) NCERT-XI, Pg#16,18

96)

SOLUTION—

Water bloom is generally caused by blue-green algae, also known as cyanobacteria. These bacteria can rapidly increase in population in aquatic environments, leading to a noticeable change in water color.

The correct answer is: OPTION(2) blue-green algae

97)

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Correct Answer is Option (1).

98) NCERT XI Page # 21

99)

Explain Question: Check whether the given examples in each column are correctly grouped into their respective kingdoms (Monera, Protista, Fungi).

Concept: This question is based on Monera, Protista, Fungi

Solution: Option (1): All groups correct.

Option (2): Chlorella is Protista, not Fungi → incorrect.

Option (3): Trichoderma & Colletotrichum are fungi, not Monera → incorrect.

Option (4): Penicillium is fungus, wrongly placed in Monera → incorrect.

Final answer: 1

100)

Correct Answer is Option (3).

101)

Generated by Allie

Problem Statement: The question asks which pollutant's higher concentration in the air can be indicated by the presence of lichens. In other words, it is about identifying the specific air pollutant that affects lichens and can be detected through their condition.

Underlying Concept: Lichens are symbiotic organisms sensitive to air quality, especially to sulfur dioxide (SOâ,,). Because of their sensitivity, they serve as bio-indicators, meaning their presence and health can indicate the level of certain air pollutants, particularly SOâ,,, as it is toxic to lichens and causes visible damage.

Tips and Tricks: Remember that lichens are especially vulnerable to sulfur dioxide (SOâ,,), so linking lichens to SOâ,, can help quickly identify the pollutant they indicate.

Common Mistakes: Confusing carbon dioxide (COâ,,) or methane (CHâ,„) as pollutants detected by lichens, while they do not directly cause damage to lichens like sulfur dioxide does.

Why Other Options Are Incorrect?: COâ,, is a greenhouse gas but does not harm lichens directly. CHâ,„ (methane) and NOâ,, (nitrogen dioxide) affect air quality but are not primary pollutants indicated by lichens.

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102)

A - *Neurospora,* B - Yeast

103)

Question Explanation: Asking about incorrect for chemosynthetic autotrophic bacteria

Concept: chemosynthetic autotrophic bacteria

Solution:

Chemosynthetic autotrophic bacteria oxidise inorganic substances only → so statement 1 (mentions organic also) is incorrect

Statement 2 → correct (ATP production from oxidation)

Statement 3 → correct (nutrient recycling)

Statement 4 → correct (they are abundant in nature)

Final answer: 1

104)

Question Explanation: Question asks which feature is NOT used by R.H. Whittaker as a basis for the five-kingdom classification system

Concept: Five Kingdom classification

Solution: Whittaker's used 5 criteria:
1. Cell type
2. Cell organization
3. Body organization
4. Mode of nutrition
5. Reproduction
He did not use reserve food materials as a criterion.

Final Answer: option (3)

105)

Explain Question: Question asks about how many given statements are not correct.

Concept: Inter of biological classification

Solution: Only statement B is incorrect because Diatom cell wall has silica, not calcium carbonate .

Final answer: Option 1

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106)

Correct Answer is Option (4).

107)

Correct Answer is Option (2).

108)

Correct Answer is Option (4).

109)

Correct Answer is Option (3).

110)

Correct Answer is Option (1).

111)

Correct Answer is Option (4).

112)

Explain Question: Question asks the causal agent of potato spindle tuber disease.

Concept: Viroids

Solution: The disease is not caused by bacteria, virus, or mycoplasma but specifically by viroids.

Final answer: 2

113)

Correct Answer is Option (4).

114)

Correct Answer is Option (4).

115) NCERT (XIth) Pg. # 24, 2.3.2

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116)

Generated by Allie

Problem Statement: The question asks us to identify which method of genetic recombination in bacteria involves the participation of a virus.

Underlying Concept: Genetic recombination in bacteria refers to the process by which bacteria exchange genetic material to produce new combinations of genes. There are several methods of genetic recombination: transformation, transduction, conjugation, and translation. Transduction is the process where viruses (bacteriophages) transfer genetic material from one bacterium to another, facilitating genetic recombination.

Tips and Tricks: Remember the root words: 'Transduction' involves 'duction' meaning to lead or bring, and it is related to viruses, whereas 'Transformation' means change by taking DNA directly from the environment.

Common Mistakes: Confusing transformation and transduction, or thinking that conjugation involves viruses. Also, translation is unrelated to genetic recombination.

Why Other Options Are Incorrect?: 'Transformation' involves bacteria taking up free DNA, not viral involvement. 'Conjugation' is direct transfer between bacterial cells using a pilus, without any virus. 'Translation' is a process in protein synthesis and does not involve genetic recombination or viruses.

117)

Explain Question: Question asks which group shows the widest metabolic diversity.

Concept: Bacteria

Solution: Among the given groups, bacteria have the most extensive metabolic diversity.

Final answer: 4

118)

Explanation :
In Whittaker's five-kingdom classification, eukaryotes are found in:

A. Protista

B. Fungi

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C. Plantae

D. Animalia

Only Monera includes prokaryotes. Hence, eukaryotes are assigned to four of the five kingdoms.
Final Answer: (3)

119)

Correct Answer is Option (1).

120)

Correct Answer is Option (4).

121) **Question Explanation:** Question is asking: Which structure is not formed during sexual reproduction in fungi.

Concept: Reproduction in fungi (General characters of fungi).

Solution: Zoospore is asexual, not sexual.

Final answer: Option (2)

122)

NCERT XI, Pg. # 128

123)

Correct Answer is Option (2).

124)

Correct Answer is Option (4).

125)

Correct Answer is Option (2).

126)

Correct Answer is Option (4).

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127)

Correct Answer is Option (4).

128) NCERT BASED

129)

Solution: The Correct answer is 2.Assertion and Reason are true and the Reason is not a correct explanation of the Assertion

130)

Correct Answer is Option (1).

131)

Correct Answer is Option (1).

132)

Correct Answer is Option (2).

133)

Question explanation: The question asks what prions are mainly made of.

Concept: Prions

Solution: Since prions contain only protein, the correct answer is protein.

Final answer: Protein

134)

Correct Answer is Option (3).

135)

Correct Answer is Option (3).

136)

Correct Answer is Option (4).

137) Ctenophores

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138) E and C

139) Porifera

140) Chondricthyes

141) Correct Answer is Option (1).

142) Only III is correct.

143) **Explain Question:** Which of these possess a 4-chambered heart?
Concept : This Question is based on Cardiac Cycle
Explanation :

Fish: 2 chambers
Amphibians (e.g., frog): 3 chambers
Most reptiles: 3 chambers (except crocodiles)
Birds and mammals: 4 chambers

Pigeons are birds, and all birds have a 4-chambered heart (2 atria and 2 ventricles) for complete separation of oxygenated and deoxygenated blood.
Final answer : Option 4: Pigeon

144) Open (arthropods→sinuses), closed (annelids/ chordates → precise regulation). Vertebrates: fish 2-chambered→amphibian/ reptile 3-chambered (mixing)→ mammal
4-chambered complete separation.

145)

A. Organ level — *Fasciola*
B. Cellular aggregate level — *Spongilla*
C. Tissue level — *Obelia*
D. Organ system level — *Pheretima*

146) In urochordates, tail is present only in larval tail cephalochordates, it extends from head to tail throughout life.

147) *Bufo, Hyla, Ichthyophis, Salamandra* and *Rana* are amphibians.
Crocodilus, Hemidactylus and *Naja* are Reptilians.

148) Tetrapoda includes Amphibia, Reptilia, Aves and Mammalia (four-limbed vertebrates).
Neophron → Aves
Ornithorhynchus → Mammalia
Hemidactylus → Reptilia

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Ichthyophis → Amphibia
All four belong to Tetrapoda, whereas other options include invertebrates or fishes.

149) **Solution :** Sponges are neither clearly diploblastic nor triploblastic; they are at cellular grade.
Final Answer : (2)

150) **Solution :** Jointed legs and exoskeleton allowed enormous diversification.
Final Answer : (1)

151) **Solution :** They have an incomplete (blind-sac) gut; metagenesis occurs in some like Obelia.
Final Answer : (2)

152) **Solution : Statement I :** The diagrammatic sectional view given is of acoelomate.
Statement II: Only Aschelminthes have pseudocoelom.
Final Answer : (4)

153) **Solution:** Both statements are correct facts about Ctenophores, but bioluminescence does not explain why they are exclusively marine.
Final Answer : (2)

154) Coelenterates (Cnidaria, e.g., Hydra, Obelia) are diploblastic
Platyhelminthes → triploblastic, acoelomate Arthropoda, Annelida → triploblastic, coelomate.

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155) Metagenesis is the alternation of sexual and asexual generations in the life cycle, e.g., in Obelia.

156)

Well developed muscular pharynx is present in Aschelminthyes.
The space between visceral hump and mantle is called mantle cavity.

157) The diagram represents Cyclostomata (e.g., Myxine, Petromyzon), which lack jaws and paired fins and have a cartilaginous skeleton.

158)

Correct Answer is Option (3).

159) **Solution :** Cnidoblast are use for anchorage, defense and for the capture of prey.
Final Answer : (2)

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160) **Solution :** The given figure is of tapeworm. Tapeworm belongs to phylum Platyhelminthes which are mostly endoparasites, bilaterally symmetrical, triploblastic and acoelomate animals with organ level of organization.
Final Answer : (4)

161) **Solution :** Locusts are the swarming phase of certain species of short-horned grasshoppers in the family Acrididae. Scorpions are predatory arthropod animals having eight legs and are easily recognised by the pair of grasping pedipalps and the narrow, segmented tail. Prawn is an arthropod (invertebrate having jointed limbs and a segmented body with an exoskeleton made of chitin. *Pila* is a genus of large freshwater snails with an operculum.
Final Answer : (1)

162) **Solution :** The given figures (A, B and C) are respectively hippocampus, catla (both are bony fishes) and salamander (amphibian). Hippocampus have a skeleton made up of bony plates, they use gills to breath and have an inflatable bladder to help regulate their buoyancy in the water. Catla is a member of ostoichthyes and possesses air bladder to regulate buoyancy.
Final Answer : (1)

163) **Explanation:** Coelenterates (Hydra) are diploblastic.
Final Answer : (3)

164) **Explanation:** Hemichordata (Balanoglossus) has proboscis, collar, trunk and pharyngeal gill slits.
Final Answer : (1)

165) **Explanation:** This is called "secondary radial symmetry," indicating they evolved from bilateral ancestors.
Final Answer : (3)

166) **Explanation:** Hemichordates possess a stomochord in the collar region, which is a short, hollow diverticulum of the gut. It is not a true notochord but resembles it superficially, distinguishing them from true chordates.
Final Answer : (2)

167) **Solution :** Hemichordata (e.g., Balanoglossus) have proboscis, collar and trunk.
Final Answer : (1)

168) **Solution :** Fertilization is mostly external; respiration by gills (larva), lungs, and skin (adult).
Final Answer : (2)

169) **Solution :** Terrestrial arachnids (scorpion) have book lungs; Limulus has book gills.
Final Answer : (2)

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170) Porifera

171) **Explanation:** Chondrichthyes (cartilaginous fishes) are characterized by having skin covered in placoid scales, also known as dermal denticles.
Final Answer : (2)

172) **Explanation:** Radula is a file-like structure in most molluscs (e.g., Pila, Chiton) for scraping food.
Final Answer : (2)

173) **Explanation:** Polyps (like Hydra) are sessile and cylindrical; Medusae (like Aurelia) are umbrella-shaped and free-swimming.
Final Answer : (1)

174) **Explanation:** Struthio (Ostrich) is the largest living bird; it has lost the ability to fly but is an exceptionally fast runner.
Final Answer : (2)

175) **Explanation:** While Platyhelminthes shows the "organ" level, Aschelminthes (roundworms) is the first to show fully developed organ systems like the digestive tract.
Final Answer : (2)

176) **Explanation:** Limulus (King Crab) has remained virtually unchanged for over 400 million years, surviving multiple mass extinctions.
Final Answer : (2)

177) **Explanation:** Trygon (Stingray) has a modified dorsal fin spine that can inject venom, whereas Torpedo (Electric ray) has electric organs.
Final Answer : (2)

178) **Explanation:** Myxine (Hagfish) is a cyclostome. They are primitive vertebrates that lack true jaws and paired fins.
Final Answer : (3)

179) Dog fish

180) **Explanation:** Annelids show true metamerism; each segment has similar organs (e.g., nephridia, ganglia).
Final Answer : (2)

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